Renal



MCQ

بسم الله الرحمن الرحيم

والصلاة والسلام على خير المرسلين محمد بن عبد الله وعلى آله وصحبه وسلم أما بعد ، نقدم بين يديك ملف أسئلة (RRS – Renal system)

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ابدأ مستعيناً بالله وكرر الحل ما استطعت واحرص على وقتك فهو رأس مالك وأغلى ما تملك

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جميع الحقوق محفوظة لفريق معوان

ولا نحلل بيعها بعائد ربحي او سرقتها باي شكل من الأشكال

Anatomy

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❖ Kidney

1-	The renal arteries originate at which of the following vertebral level?		
A-	T12	B-	L1
C-	L2	D-	L3
E-	L4		

2-	Regarding the left renal vein, which of the following statement is correct?
A-	Is crossed anteriorly by the inferior mesenteric artery
B-	Lies behind the left renal artery
C-	Joins the portal vein behind the pancreas
D-	Receives the inferior mesenteric vein

3-	Which of the following is NOT related to the anterior surface of the right kidney?		
A-	The 2nd part of the duodenum	B-	The right lobe of liver
C-	The right suprarenal gland	D-	Duodenojejunal flexure

4-	Which of the following is NOT true about relations of the kidney?
A-	The psoas major muscle posteriorly
B-	The second part of the duodenum, anterior to the right kidney
C-	The tail of the pancreas, anterior to the left kidney
D-	The stomach anterior to the left kidney
E-	The splenic artery, anterior to the left kidney

5-	Which of the following is NOT true about the kidney?
A-	The hilum of the left kidney is just below that of the right
B-	The hilum faces medially
C-	The long axis is parallel with the lateral border of psoas major
D-	The perinephric fat is surrounded by the renal fascia
E-	The renal artery lies behind the renal vein

6-	Which of the following is NOT a branch of the lumbar plexus?		
A-	The iliohypogastric nerve	B-	The genitofemoral nerve
C-	The femoral nerve	D-	The obturator nerve
E-	The pudendal nerve		

﴿وَاتْلُ مَا أُوحِيَ إِلَيْكَ مِن كِتَابِ رَبِّكَ لَا مُبَدِّلَ لِكَلِمَاتِهِ وَلَن تَجِدَ مِن دُونِهِ مُلْتَحَدًا﴾

7-	**Which of the following overlies the superior extremity of kidney?**		
A-	Diaphragm	B-	2nd part of duodenum
C-	Head of pancreas	D-	Suprarenal gland
E-	Pylorus		

8-	**Which of the following is a branch of the lumbar plexus?**		
A-	Sciatic nerve	B-	Inferior gluteal nerve
C-	Superior gluteal nerve	D-	Obturator nerve
E-	Pudendal nerve		

9-	**Regarding the structures passing through the hilum of the kidney. Which is true?**		
A-	Renal v.; anteriorly	B-	Renal a,; in the middle
C-	Renal pelvis; posteriorly	D-	Lymphatic vessels
E-	All of the above		

10-	**The coverings of the kidney include all of the following EXCEPT which one?**		
A-	Fibrous capsule	B-	Perirenal fat
C-	Renal fascia	D-	Pararenal fat
E-	Peritoneum		

11-	**Which of the following is NOT related to the anterior surface of the right kidney?**		
A-	Right lobe of liver	B-	Right colic flexure
C-	2nd part of duodenum	D-	Stomach
E-	Right suprarenal gland		

12-	**Which of the following is NOT related to the anterior surface of the left kidney?**		
A-	Stomach	B-	Body of pancreas
C-	Transverse colon	D-	Left colic flexure
E-	Descending colon		

13-	**While exposing the kidney from behind, all of the following nerves are liable to injury except which?**		
A-	Lateral cutaneous nerve of the thigh	B-	Ilioinguinal nerve
C-	Subcostal nerve	D-	Iliohypogastric nerve

14-	**Inferiorly, Which of the following is NOT related to the posterior surfaces of the kidney?**		
A-	Subcostal nerve	B-	Iliohypogastric nerve
C-	Ilioinguinal nerve	D-	2nd portion of the duodenum
E-	Quadratuslumborum muscle		

15-	**All of the following statements concerning the hilum of the kidney are correct EXCEPT?**
A-	It is located on the concave medial margin of the kidney
B-	It is the site of entrance of the structures into and from the kidney
C-	The vein infront artery
D-	The artery behind the renal pelvis
E-	The renal pelvis is directed downwards

16-	**Which of the following structures is related to the spleen, stomach, pancreas, and the left crus of the diaphragm?**		
A-	Left suprarenal gland	B-	Left kidney
C-	Left gonadal vein	D-	Transverse mesocolon
E-	Abdominal aorta		

17-	**One of the following structures lies behind the left kidney?**		
A-	The splenic vessels	B-	The last rib
C-	The stomach	D-	The left colic flexure

18-	**The right kidney is NOT related anteriorly?**		
A-	Right lobe of liver	B-	Right supra-renal gland
C-	Right colic flexure	D-	Right gonadal artery

19-	**The left kidney is not related anteriorly to one of the following?**		
A-	Supra-renal gland	B-	Stomach
C-	Spleen	D-	Liver

20-	**Regarding the kidney, one of the following statements is correct?**
A-	The right kidney is slightly higher than the left kidney
B-	The kidney is divided into five renovascular segments
C-	Ath the hilum of the kidney, the renal artery lies anterior to the renal vein
D-	The kidney and the suprarenal gland lie together within the same sheath of perirenal fascia

21-	**In the human kidney, the renal papilla projects directly into the?**		
A-	Renal pyramid	B-	Ureter
C-	Major calyx	D-	Renal columns
E-	Minor calyx		

فَالعِلْمُ حِلْيَةُ قَوْمٍ لَا حُلِيَّ لَهُمْ *** وَالجَهْلُ ذِئْرٌ وَبِئْسَ الثَّوْبُ لِلْجَسَدِ

22-	**The following are related to the anterolateral surface of the right kidney?**		
A-	Body of pancreas	B-	Second part of duodenum
C-	Right lobe of liver	D-	Right suprarenal gland
E-	Coils of the small intestine		

23-	**The posterior relations of kidney include the following EXCEPT?**		
A-	Psoas major muscle	B-	Diaphragm
C-	Internal oblique muscle	D-	Quadratus lumborum muscle
E-	Transversus abdominis muscle		

24-	**The covering of the kidney includes the following EXCEPT?**		
A-	Fibrous capsule	B-	Thoracolumbar fascia
C-	Perirenal fat	D-	Renal fascia
E-	Pararenal fat		

25-	**Which structure belongs to renal cortex?**		
A-	Minor renal calices	B-	Renal column
C-	Renal papillae	D-	Renal pelvis
E-	Renal pyramid		

26-	**The upper pole of kidney is in neighborhood of?**		
A-	Diaphragm	B-	Superior portion of duodenum
C-	Head of pancreas	D-	Suprarenal gland
E-	Pylorus		

27-	**The following structures lie in front of right kidney EXCEPT?**		
A-	Right lobe of liver	B-	Right colic flexure
C-	Second part of duodenum	D-	Right suprarenal gland
E-	Pancreas		

28-	**The following structures lie in front of left kidney EXCEPT?**		
A-	Stomach	B-	Pancreas
C-	Jejunum	D-	Left colic flexure
E-	Second part of duodenum		

29-	**An organ or structure that is not a component of the urinary system is the?**		
A-	Urethra	B-	Urinary bladder
C-	Ureter	D-	Adrenal gland
E-	Kidney		

30-	**Which of the following statements concerning the kidneys is false?**
A-	They are retroperitoneal
B-	They each contain 8 to 15 renal pyramids
C-	They each have a distinct renal cortex and renal medulla region
D-	They are positioned between the third and fifth lumbar
E-	They developed from intermediate mesoderm

31-	**Which of the following is the correct sequence in which urine flows through the kidney toward the urinary bladder?**
A-	Renal pelvis, major calyx, minor calyx, papillary duct, ureter
B-	Papillary duct, minor calyx, major calyx, renal pelvis, ureter
C-	Minor calyx, major calyx, papillary duct, renal pelvis, ureter
D-	Papillary duct, major calyx, minor calyx, ureter, renal pelvis

32-	**The apex of the renal pyramid is called the?**		
A-	Mahor calyx	B-	Minor calyx
C-	Renal papilla	D-	Renal pelvis
E-	Ureter		

33-	**Inferiorly, the posterior surfaces of the kidney are related to all of the following structures EXCEPT?**		
A-	Subcostal nerve	B-	Iliohypogastric nerve
C-	Ilioinguinal nerve	D-	2nd portion of the duodenum
E-	Quadratus lumborum		

34-	**The renal papillae empty into which of the following?**		
A-	Renal vein	B-	Ureter
C-	Minor calyces	D-	Renal pyramid
E-	Renal column		

35-	**All of the following statements concerning the renal hilum are correct EXCEPT?**
A-	The renal vein is anterior to the renal artery
B-	The renal artery is anterior to the renal pelvis
C-	It is the entrance to the renal sinus
D-	It is located on the concave medial margin of the kidney
E-	It contains the renal pyramids

36-	The nerves of the kidneys and suprarenal glands are derived from which of the following plexuses?		
A-	Celiac	B-	Lumbar
C-	Inferior mesenteric	D-	Sacral
E-	Inferior hypogastric		

37-	The renal pyramids are located within the?		
A-	Column	B-	Cortex
C-	Medulla	D-	Pelvis
E-	Ureter		

38-	Each minor calyx receives urine from the?		
A-	Renal papillae	B-	Pelvis
C-	Ureter	D-	Columns
E-	Mahor calyx		

يقول الإمام الغزالي – رحمه الله :–

اعلم ان للدين شطرين، أحدهما: ترك المناهي، والآخر: فعل الطاعات . . وترك المناهي هو الأشد؛ فإن الطاعات يقدر عليها كل واحد، وترك الشهوات لا يقدر عليه إلا الصديقون، فلذلك قال رسول الله صلى الله عليه وسلم: (المهاجر من هجر السوء، والمجاهد من جاهد هواه).

واعلم أنك إنما تعصي الله بجوارحك، وهي نعمة من الله عليك وأمانة لديك، فاستعانتك بنعمة الله على معصيته غاية الكفران، وخيانتك في أمانة استودعها الله غاية الطغيان؛ فأعضاؤك رعاياك، فانظر كيف ترعاها؛ فكلكم راع، وكلكم مسؤول عن رعيته .

❖ Ureter

39-	Ureteric constriction is seen at all of the following positions EXCEPT which one?		
A-	Pelvi-ureteric junction	B-	Crossing of iliac artery
C-	Uretero-vesicle junction	D-	Ischial spine

40-	Which of the following is NOT true about the ureters?
A-	Muscular tubes
B-	Each emerges from the hilum in between the renal vessels
C-	Continuous superiorly with the funnel-shaped renal pelvis
D-	Continuous inferiorly with the urinary bladder
E-	Each is 25 cm in length

41-	Regarding the ureter, one of the following statements is NOT correct?
A-	It is about 25 cm long
B-	It descends vertically along the tips of transverse processes of L2-L5
C-	It is crossed in the abdomen by the gonadal artery
D-	The inferior mesenteric vein lies on the lateral side of the left ureter

42-	Regarding the ureter, one of the following statements is correct?
A-	Pain sensations is carried from the ureter to the upper two thoracic and the upper two lumbar segments of the spinal cord
B-	In the female, it is crossed by the vaginal artery just lateral to the cervix of the uterus
C-	While lying on the psoas major, it is crossed by the genito-femoral nerve
D-	It crosses the bifurcation of the common iliac artery

43-	The ureter is supplied by the following arteries EXCEPT?		
A-	Renal artery	B-	Abdominal aorta
C-	Gonadal artery	D-	Splenic artery
E-	Common iliac artery		

44-	The female ureter crosses?		
A-	Above uterine artery	B-	Under uterine artery
C-	In front of uterine artery	D-	Inside uterine artery
E-	Lateral to uterine artery		

ما أبعد الغايات إلا أنني *** أجد الثبات لكم بهن كفيلا

فكلوا إلى الله النجاح وثابروا *** فالله خير كافلا ووكيلا

45-	**The following are FALSE about the pelvic part of ureter EXCEPT?**
A-	Descends immediately behind the internal iliac artery
B-	Crosses lateral to the external vessels, umbilical artery, obturator nerve and vessels
C-	Crosses posteriorly over the ductus deferens (male)
D-	Passes lateral to the fornex of the vagina and is crossed by uterine artery
E-	Ends on the inferior of the urinary bladder as the internal urethral orifice

46-	**The structure that connects a kidney to the urinary bladder is the?**		
A-	Ureter	B-	Urethra
C-	Renal pelvis	D-	Collecting duct
E-	Glomerulus		

47-	**All of the following statements concerning the ureters are correct EXCEPT?**
A-	They are retroperitoneal
B-	They cross the pelvic brim anterior to the bifurcation of the common iliac arteries
C-	They are fibrous tubes connecting the kidneys to the urinary bladder
D-	Their superior halves lie in the abdomen and the inferior halves lie in the pelvis
E-	Their oblique passage through the bladder wall forms a one-way “flap valve”

عن أبي هريرة – رضي الله عنه – أنَّ رَسولَ اللَّهِ صَلَّى اللهُ عليه وسلَّمَ، قالَ:

(مَن أنْفَقَ زَوْجَيْنِ في سَبيلِ اللَّهِ، نُودِيَ مِن أبْوابِ الجَنَّةِ: يا عَبْدَ اللَّهِ هذا خَيْرٌ، فمَن كانَ مِن أهْلِ الصَّلَاةِ دُعِيَ مِن بَابِ الصَّلَاةِ، ومَن كانَ مِن أهْلِ الجِهَادِ دُعِيَ مِن بَابِ الجِهَادِ، ومَن كانَ مِن أهْلِ الصِّيَامِ دُعِيَ مِن بَابِ الرَّيَّانِ، ومَن كانَ مِن أهْلِ الصَّدَقَةِ دُعِيَ مِن بَابِ الصَّدَقَةِ، فقالَ أبو بَكْرٍ رَضِيَ اللَّهُ عنْه: بأَبِي أنْتَ وأُمِّي يا رَسولَ اللَّهِ ما علَى مَن دُعِيَ مِن تلكَ الأبْوابِ مِن ضَرُورَةٍ، فَهلْ يُدْعَى أحَدٌ مِن تلكَ الأبْوابِ كُلِّهَا، قالَ: نَعَمْ وأَرْجُو أنْ تَكُونَ منهمْ).

❖ Urinary bladder

48-	The neck of the bladder in females is held firmly by which of the following structures?		
A-	Puboprostatic ligaments	B-	Puborectalis muscle
C-	Levatorani muscle	D-	Coccygeus muscle
E-	Pubovesical ligaments		

49-	The median umbilical ligament contains which of the following structures?		
A-	Inferior epigastric vessels	B-	Obturator vessels
C-	Remnant of the urachus	D-	Uterine tubes
E-	Ovarian vessels		

50-	All are true about the trigone of the urinary bladder EXCEPT which of the followings?
A-	Mucosa is tightly adherent to the underlying musculature
B-	Mucosa is smooth
C-	It is lined by transitional epithelium
D-	It is derived from the urogenital sinus

51-	Trigone of urinary bladder surrounded by all of the followings EXCEPT which?		
A-	Left ureteral orifice	B-	Right ureteral orifice
C-	Internal urethral orifice	D-	Apex of bladder

52-	All are true about the trigone of the urinary bladder EXCEPT which?
A-	In the empty state its mucosa is thrown into many folds
B-	Bounded by ureteric orifices
C-	The internal urethral orifice forms its apex
D-	It is derived from the absorbed part of the mesonephric duct

53-	Posterior relations of the urinary bladder in female not include which of the followings?		
A-	Body of uterus	B-	Uterovesical pouch
C-	Loops of small intestine	D-	Vagina

54-	The following are FALSE about urinary bladder EXCEPT?
A-	Is a pelvic organ in the infant
B-	Has peritoneum on its anterior surface when full
C-	Is firm Musculo-glandular organ
D-	Its muscular walls are pierced obliquely by the ureters
E-	The urachus is a safety valve allowing overflow when the urethra is obstructed

55-	**The trigone of bladder lies?**		
A-	IN internal surface of fundus of bladder	B-	Below apex of urinary bladder
C-	Body of bladder	D-	Below neck of bladder
E-	In base of prostate		

56-	**The following are false about urinary bladder EXCEPT?**
A-	May rise into the peritoneal cavity when it is filled with urine
B-	Contacts with prostate posteriorly (male)
C-	Contacts with posterior part of fornix (female)
D-	Contacts with the urogenital diaphragm at the apex
E-	Is entirely within the lesser pelvis when it is empty

57-	**The mark to find the ureteric orifice in urinary bladder is?**		
A-	Trigone of bladder	B-	Internal urethral orifice
C-	Interureteric fold	D-	Urethral ridge
E-	Vesical uvula		

58-	**The uvula of the bladder is located in which of the following locations?**		
A-	Retropubic space	B-	Trigone of the bladder
C-	Apex of the bladder	D-	Fundus of the bladder
E-	Pubovesical ligament		

59-	**Parasympathetic fibers to the bladder are derived from which of the following nerves?**		
A-	Pelvic splanchnic	B-	Greater splanchnic
C-	T11-L2	D-	Superior hypogastric plexus
E-	Sacral plexus		

60-	**Which of the following structures separates the bladder from the pubic bones in females?**		
A-	Rectouterine fold	B-	Vesicouterine pouch
C-	Trigone of the bladder	D-	Median umbilical fold
E-	Retropubic space		

كَانَ أَبُو بَكْرٍ إِذَا أَخَذَتْهُ الحُمَّى يَقُولُ:

كُلُّ امْرِئٍ مُصَبَّحٌ فِي أَهْلِهِ . . . وَالمَوْتُ أَدْنَى مِن شِرَاكِ نَعْلِهِ

❖ Urethra

61-	**Prostatic urethra is characterized by all of the following EXCEPT which?**
A-	The narrowest and least dilatable part of male urethra
B-	Presents a posterior median crest
C-	Lies closer to anterior surface of prostate
D-	Receives ducts of the prostatic glands
E-	The seminal colliculus is a dilatation near its middle

62-	**Which of the following is the shortest part of make urethra?**		
A-	Prostatic	B-	Membranous
C-	Bulbar	D-	Penile

63-	**Spongy part of the male urethra drains via which lymph nodes?**		
A-	Superficial inguinal lymph nodes	B-	External Iliac lymph nodes
C-	Deep inguinal lymph nodes	D-	Aortic lymph nodes
E-	Paraaortic lymph nodes		

64-	**Which of the following statements best explains the external urethral sphincter?**
A-	Smooth muscle
B-	Innervated by the perineal nerve
C-	Lying between the perineal membrane and Colles' fascia
D-	Part of the pelvic diaphragm
E-	It is formed of two parts in females

65-	**The narrowest part of male urethra is?**		
A-	Prostatic part	B-	Membranous part
C-	External urethral meatus	D-	Internal urethral orifice
E-	Spongy part		

66-	**In the male the membranous urethra?**		
A-	Is that part within the prostate gland	B-	Starts at the trigone of the bladder
C-	Is that part within the urogenital diaphragm	D-	Is that part within the corpus spongiosum
E-	Is called the ejaculatory duct		

ذو العقل يشقى في النعيم بعقله *** وأخو الجهالة في الشقاوة ينعم

67-	**The female urethra is?**		
A-	6 cm long	B-	8 cm long
C-	4 cm long	D-	10 cm long
E-	20 cm long		

68-	**The vas deferens is connected to the prostatic urethra by?**		
A-	The prostatic utricle	B-	Gartner's duct
C-	The ejaculatory duct	D-	The urachus
E-	None of the above		

69-	**Second constriction of male urethra lies in?**		
A-	Prostatic portion of urethra	B-	Membranous part
C-	Cavernous part	D-	External orifice of urethra
E-	Internal urethral orifice		

70-	**All of the following statements are true related to the urethra of a male EXCEPT?**
A-	It serves both the urinary and reproductive systems
B-	It contains a single urethral sphincter near the neck of the urinary bladder
C-	It receives secretions from the bulbourethral glands
D-	It consists of three distinct regions
E-	It is approximately 20 cm long in an adult

71-	**Which of the following parts of the male urethra is the widest and most dilatable?**		
A-	Preprostatic	B-	Prostatic
C-	Membranous	D-	Spongy
E-	External urethral meatus		

72-	**Which of the following structures opens into the prostatic sinus?**		
A-	Prostatic utricle	B-	Ejaculatory ducts
C-	Prostatic ductulus	D-	Seminal vesicles
E-	Bulbourethral glands		

73-	**The paraurethral glands open into which of the following structures?**		
A-	Near the external urethral orifice	B-	In the neck of the bladder
C-	In the prostatic utricle	D-	In the seminal colliculus
E-	In the ejaculatory ducts		

74-	The navicular fossa is located in which of the following structures?		
A-	Glans penis	B-	Prostate
C-	Bladder	D-	Seminal vesicle
E-	Expanded in the clitoris		

75-	The membranous part of the male urethra is located in which of the following structures?		
A-	Bladder	B-	Prostate
C-	External urethral sphincter	D-	Bulb of penis
E-	Glans penis		

76-	The paraurethral glands are homologues to which of the following structures?		
A-	Testes	B-	Prostate
C-	Seminal vesicles	D-	Bulbourethral glands
E-	Epididymis		

77-	Male urethra is approximately _____ long?		
A-	30 cm – 35 cm	B-	15 cm – 20 cm
C-	10 cm – 15 cm	D-	5 cm – 10 cm
E-	25 cm – 30 cm		

يقول ابن القيم – رحمه الله تعالى –:

فما أذنب عبد ذنبًا إلا زالت عنه نعمة من الله بحسب ذلك الذنب، فإن تاب وراجع رجعت إليه أو مثلها، وإن أصر لم ترجع إليه، ولا تزال الذنوب تزيل عنه نعمه حتى تسلب النعم كلها، قال الله تعالى: ﴿إِنَّ اللَّهَ لَا يُغَيِّرُ مَا بِقَوْمٍ حَتَّىٰ يُغَيِّرُوا مَا بِأَنفُسِهِمْ﴾ [الرعد: ١١]

❖ Embryology

78-	**The collecting system (collecting tubules), major and minor calyces, renal pelvis and ureter are all derived from which embryonic structure?**		
A-	Pronephric tubules	B-	Metanephric cap
C-	Mesonephric tubules	D-	Ureteric bud

79-	**In the adult kidney, the excretory system (nephrons) is derived from which embryonic structure?**		
A-	Ureteric bud	B-	Metanephric cap
C-	Mesonephric duct	D-	Paramesonephric duct

80-	**Which statement is true in regards to the pronephros?**
A-	Develops in the thoracic and lumbar regions of the nephrogenic cord
B-	Forms 70—80 pairs of pronephric tubules
C-	Pronephric duct persists and later continues as mesonephric duct
D-	Pronephric tubules are functional in mammals

81-	**Which abnormality is correctly paired with its description?**
A-	Polystic kidney – communication between the developing nephron and collecting tubules
B-	Ectopic ureter – ureter opens into the urinary bladder
C-	Horse-shoe kidney – two kidneys may fuse
D-	Renal agenesis – presence of an extra kidney

82-	**In males, mesonephric tubules give rise to?**		
A-	Head of epididymis	B-	Body & tail of epididymis
C-	Vas deference	D-	Ejaculatory duct

83-	**In females, mesonephric tubules give rise to?**		
A-	Garnter's duct	B-	Epoophron & paroophron
C-	Fallopian tube	D-	Vagina

84-	**The trigone of urinary bladder is derived from?**		
A-	Vescourethral canal	B-	Mesonephric duct
C-	Paramesonephric duct	D-	Vetelline duct

85-	**Which structure is responsible for dividing the cloaca?**		
A-	Tracheoesophageal septum	B-	Urorectal septum
C-	Conotruncal septum	D-	Septum in the AV canal

86-	**The proximal part of allantois forms?**		
A-	Trigone of urinary bladder	B-	Superior surface of urinary bladder
C-	Neck of urinary bladder	D-	Apex of urinary bladder

87-	**The distal portion of the allantois degenerates into which structure?**		
A-	Calyces	B-	Urachus
C-	Ureter	D-	Urethra

88-	**Hypospadius means?**
A-	The urethra does not open
B-	The urethra opens on the under surface of the penis
C-	The urethra opens on the dorsum of the penis
D-	The urethra opens on the abdominal wall

89-	**In males, seminiferous tubules develope from?**		
A-	Medullary cords	B-	Mesonephric tubules
C-	Sertoli cells	D-	Talbuginea

90-	**Congenital hydrocele means?**
A-	Persistence of the whole perocessus vaginalis
B-	Obliteration of the whole perocessus vaginalis
C-	Obliteration of the proximal part of perocessus
D-	Obliteration of the distal part of perocessus vaginalis

91-	**The testis descends into the scrotum during the?**		
A-	3rd month	B-	6th month
C-	7th month	D-	9th month

92-	**Primordial germ cells are derived from?**		
A-	Intermediate mesoderm	B-	Coelmic epithelium
C-	Endoderm of the yolk sac	D-	Paraaxial mesoderm

93-	**The following structure is derived from the ureteric bud?**		
A-	Major calyces	B-	Bowman's capsule
C-	Distal convoluted tubules	D-	Proximal convoluted tubules

94-	**Regarding the development of the urinary bladder and urethra, mark one correct statement?**
A-	Epithelium of the urinary bladder is mesodermal in origin
B-	Trigone of the urinary bladder is formed by absorption of the paramesonephric ducts
C-	Pelvic part of urogenital sinus is male gives rise to membranous urethra and lower part of the prostatic urethra
D-	The lowermost part of the male urethra is mesodermal in origin

95-	**The following structure develops from the mesonephric duct in male?**		
A-	Paradidymis	B-	Vasa efferentia
C-	Seminal vesicle	D-	Prostatic utricle

96-	**Which of the following structures is a derivative of the mesonephric duct?**		
A-	Ureter	B-	Loop of henle
C-	Bowman's capsule	D-	Appendix of testis

97-	**Regarding the development of the kidney, mark one correct statement?**
A-	The collecting systems of metanephros develops from the metanephric cap
B-	The excretory system of metanephros develops from the ureteric bud
C-	Early splitting of the ureteric bud results in duplication of the ureter
D-	The ureteric bud arises from the paramesonephric duct

98-	**The following part is derived from the metanephric cap?**		
A-	Bowman's capsule	B-	Minor calyx
C-	Pelvis of ureter	D-	Collecting duct

99-	**Regarding development of the kidney, mark one correct statement?**
A-	The mesonephric ducts open cranially in the coelomic cavity
B-	The kidney is lobulated in its early stages of development
C-	The renal artery from the abdominal aorta supplies the kidney in all stages of its development
D-	In early stages of development of the kidney, its hilum is directed dorsally

100-	**Which of the following gives rise to all structures of the kidney?**		
A-	Somitic mesoderm	B-	Intermediate mesoderm
C-	Splanchnic lateral plate mesoderm	D-	Somatic lateral plate mesoderm
E-	Neural crest		

﴿الَّذِينَ آمَنُوا وَتَطْمَئِنُّ قُلُوبُهُمْ بِذِكْرِ اللَّهِ أَلَا بِذِكْرِ اللَّهِ تَطْمَئِنُّ الْقُلُوبُ﴾

101-	**The following is WRONG about ureteric bud?**		
A-	Endodermal derivative	B-	Arises from Wolffian duct
C-	Forms collecting tubules	D-	Penetrates metanephric tissue
E-	Present in both males and females		

102-	**The urinary system develops from?**		
A-	Lateral plate mesoderm	B-	Endoderm
C-	Splanchnic mesoderm	D-	Intermediate mesoderm
E-	Ectoderm		

103-	**The urachus is a remnant of?**		
A-	Urogenital sinus	B-	Urogenital ridge
C-	Allantois	D-	Cloaca
E-	Mesonephric duct		

104-	**The urinary bladder (except the trigone) is derived from tissue associated with the?**		
A-	Metanephric blastema	B-	Mesonephric ducts
C-	Paramesonephric ducts	D-	Cloaca / urogenital sinus
E-	Paraaxial mesoderm		

105-	**The glomeruli of the kidney are derived from the?**		
A-	Ureteric bud	B-	Metanephric blastema
C-	Paramesonephric ducts	D-	Mesonephric ducts
E-	Cloaca		

106-	**The developing kidneys can sometimes fuse into a single, horseshoe-shaped organ that will usually be found?**
A-	In the pelvic cavity, wrapped around a right or left common iliac artery
B-	In the abdominal cavity, at the level of the inferior mesenteric artery
C-	In the abdominal cavity, at the level of the superior mesenteric artery
D-	In the abdominal cavity, at the level of the celiac artery
E-	In its normal location in the abdominal cavity, but almost always with supernumerary arteries

107-	**The ductus deferens is derived from the?**		
A-	Ureteric bud	B-	Metanephric blastema
C-	Paramesonephric ducts	D-	Mesonephric ducts
E-	Allantois		

108-	**The collecting tubules of the kidney are derived from the?**		
A-	Ureteric bud	B-	Metanephric blastema
C-	Mesonephric ducts	D-	Paramesonephric duct
E-	Allantois		

ANSWERS – Anatomy									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	C	D	C	A	E	D	D	E	E
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	C	A	D	D	B	B	D	D	B
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
E	A	C	B	B	D	E	E	D	D
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
B	C	D	C	E	A	C	A	B	B
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
D	D	D	A	D	A	C	E	C	D
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
D	A	C	A	A	E	C	B	A	E
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**
A	B	C	B	B	C	C	C	B	B
71	**72**	**73**	**74**	**75**	**76**	**77**	**78**	**79**	**80**
B	C	A	A	C	B	B	D	B	C
81	**82**	**83**	**84**	**85**	**86**	**87**	**88**	**89**	**90**
C	B	B	B	B	D	B	B	B	A
91	**92**	**93**	**94**	**95**	**96**	**97**	**98**	**99**	**100**
D	C	A	C	C	A	C	A	B	B
101	**102**	**103**	**104**	**105**	**106**	**107**	**108**	**109**	**110**
A	D	C	D	B	B	D	A	-	-

سبحان الله وبحمده سبحان الله العظيم

Histology

❖ Histology

1-	**Which structure begins at urinary pole of renal corpuscle?**
A-	Renal artery and renal vein
B-	Peritubular capillary
C-	Proximal convoluted tubule and loop of Henle
D-	Glomerulus

2-	**Where is loop of Henle?**
A-	Between renal artery and renal vein
B-	Between peritubular capillary and collecting duct
C-	Between proximal and distal convoluted tubules
D-	Between afferent and efferent arterioles

3-	**The outermost covering of the kidney is the?**		
A-	Cortex	B-	Medulla
C-	Pelvis	D-	Capsule

4-	**Which of the following is true regarding renal lobe?**
A-	Is separated from other lobes by connective tissue septa
B-	Is visible grossly as a rounded bulge on the kidney surface
C-	Consists of one medullary pyramid together with the associated cortex
D-	Consists of one medullary ray and its associated cortex
E-	All of the above

5-	**Which region of the kidney contain glomeruli?**		
A-	Cortex	B-	Medulla
C-	Calyx	D-	pelvis
E-	Papillae		

6-	**The end of each medullary pyramid through which collecting ducts open into a calyx is called a ……..?**		
A-	Medullary ray	B-	Renal lobule
C-	Column of Bertin	D-	Renal papilla
E-	Hilum		

7-	**Regarding the lining epithelium of urinary bladder, which one is CORRECT?**		
A-	Simple columnar	B-	Transitional
C-	Stratified squamous	D-	Simple cuboidal

8-	**What is the principal function of the juxtaglomerular apparatus?**
A-	Filtration of blood from capillaries into the urinary space
B-	Active resorption of nutrients from ultrafiltrate back into the blood
C-	Volume reduction of tubular fluid, with ions pumped from lumen into intersitium followed by diffusion of water
D-	Regulated passive resorption of water through the medullary concentration gradient
E-	Regulation of plasma volume and blood pressure through renin secretion

9-	**Cells located in matrix between glomerular capillaries are called?**		
A-	Pericytes	B-	Juxtaglomerular cells
C-	Podocytes	D-	Mesangial cells
E-	Fibroblasts		

10-	**Where is extraglomerular mesangial cells (lacis cells) located?**		
A-	At the vascular pole of the glomerulus	B-	At the urinary pole of the glomerulus
C-	In between the distal tubule lining cells	D-	In between the proximal tubule lining cells
E-	Between adjacent podocyte foot processes		

11-	**Glomerular capillaries are supported on the outside (adjacent to the urinary space by)?**		
A-	Juxtaglomerular cells	B-	Fibroblasts
C-	Podocytes foot processes	D-	Pericytes
E-	Macula densa cells		

12-	**Bundles of parallel vessels which extend into the medulla are called?**		
A-	Interlobar arteries	B-	Interlobular arteries
C-	Arcuate arteries	D-	Afferent arteries
E-	Vasa recta		

13-	**Vessels which connect interlobular arteries with glomerular capillaries are called?**		
A-	Arcuate arteries	B-	Afferent arterioles
C-	Efferent arterioles	D-	Peritubular capillaries
E-	Vasa recta		

14-	**One of the following not present in kidney cortex?**		
A-	Renal corpuscles	B-	Cortical labyrinth
C-	Medullary rays	D-	Peritubular capillaries
E-	Ducts of Bellini		

15-	**Renal column of bertin is?**
A-	the pyramid and its surrounding tissues
B-	an extension of cortex between adjacent pyramids
C-	Beginning of minor calyx
D-	the 1st part of collecting tubules
E-	3rd part of collecting tubules

16-	**As regard renal column of Bertin, only one is true?**		
A-	Terminal part of collecting tubules	B-	Terminal part of DCT
C-	Terminal part of the ureter	D-	Terminal part of PCT
E-	An extension of cortical tissue into medulla		

17-	**One of the following is not a component of the nephron?**		
A-	Malpighian corpuscle	B-	Loop of Henle
C-	Ducts of Bellini	D-	Proximal convoluted tubules
E-	Distal convoluted tubules		

18-	**One of the following about the nephron is wrong?**		
A-	Functional and structural unit of the kidney	B-	About 1.5 million nephrons in each kidney
C-	Originates from the metanephrons	D-	Represented by renal corpuscle only
E-	Two types, cortical and juxtamedullary nephrons		

19-	**All the following about renal corpuscle are true except one?**		
A-	They are rounded bodies in renal cortex	B-	Their diameter 150- 250 um
C-	Surrounded by single layered capsule	D-	Have two poles; vascular pole and urinary pole
E-	Contain glomerular capillaries and mesangium		

20-	**The renal corpuscle does not contain one of the following?**		
A-	Podocytes	B-	Mesangeal cells
C-	Tuff of capillaries	D-	Macula densa
E-	Parietal cells of Bowman capsule		

21-	**One of the following about renal corpuscle is wrong?**		
A-	They are present in the renal medulla	B-	It is part of the nephron
C-	Their diameter 150-250 um	D-	Surrounded by Bowman's capsule
E-	Contain capillaries and messangium		

22-	**Parietal layer of Bowman's capsule is formed of?**		
A-	Simple squamous epithelium	B-	Simple cubical epithelium
C-	Reticular C.T only	D-	Dense C.T only
E-	Simple squamous epithelium surrounded by Reticular C.T		

23-	**Which of the following cell represent visceral layer of Bowman's capsule?**		
A-	Continuous endothelium	B-	Podocytes
C-	Cubical ciliated epithelium	D-	Fenestrated endothelium
E-	Cubical epithelium without microvilli		

24-	**Only one about the podocytes is wrong?**		
A-	Modified simple squamous cells	B-	Have acidophilic cytoplasm
C-	Secrete renin	D-	Terminate on the blood capillaries by pedicles
E-	Share in blood renal barrier		

25-	**Urinary pole of the renal capsule is connected with?**		
A-	Proximal convoluted tubules	B-	Collecting ducts
C-	Descending thin limb of loop of Henle	D-	Distal convoluted tubules
E-	Ascending thin limb of loop of Henle		

26-	**All the following about the podocytes are true except one?**		
A-	They are parietal layer of Bowman's capsule	B-	Stellate in shape with acidophilic cytoplasm
C-	Have several processes (major & minor)	D-	Contractile cells
E-	Share in blood renal barrier		

27-	**As regard podocytes, one is incorrect?**
A-	On external surface of glomerular capillaries
B-	Formed of body and multiple processes
C-	from the glomerular capillaries by subpodocytic space
D-	Their pedicles have a well-developed glycocalyx
E-	They are supporting and phagocytic cells

28-	**The glomerular capillaries are lined by?**
A-	Simple squamous fenestrated endothelial cells without diaphragm
B-	Simple squamous fenestrated endothelial cells with diaphragm
C-	Simple squamous continuous endothelial cells
D-	Intraglomerular Mesangeal cells
E-	Podocytes

29-	**One of the followings is not component of blood renal barrier?**
A-	Basal lamina of podocytes
B-	Mesangeal cells
C-	Basal lamina of endothelial cells
D-	Endothelial cells of glomerular capillaries
E-	Pedicles of podocytes and the diaphragm cover the filtration slits

30-	**All the following about PCT are true except one?**		
A-	About 15mm in length and 60 um in diameter	B-	Starts at the urinary pole of the renal corpuscle
C-	Have convoluted part and straight part	D-	Ends in the straight descending loop of Henle
E-	The Target part of aldosterone		

31-	**It is pericyte like cells present between and outside the capillary loops of renal corpuscle in non podocytic areas" identifies the cells?**		
A-	Intraglomerular messangial cells	B-	Bowman's cells
C-	Extraglomerular messangial cells	D-	Facet cells
E-	Bertin cells		

32-	**One of the following not functions of intraglomerular messangial cells?**
A-	Secretion of collagen (IV) and extracellular matrix
B-	Phagocytes of foreign bodies in glomerular blood or in intercapillary spaces
C-	Supporting function
D-	Contractile (has receptors for angiotensin II
E-	Share in blood renal barrier

33-	**Filtration slits present between?**
A-	Pedicles of minor processes of podocyte
B-	Major process of podocyte
C-	Major calyx
D-	Minor calyx
E-	Parietal and visceral layers of Bowman's capsule

34-	**As regard PCT, one statement is wrong?**
A-	Lined by simple cubical epithelium
B-	Cells have acidophilic brush border
C-	Have clear cell borders
D-	Have acidophilic basal striation
E-	Number of cells is little when compared with DCT

35-	**The main ultra-structure features of the cells lining DCT are?**		
A-	Numerous lysosomes	B-	Basal infolding
C-	Well developed SER	D-	Numerous RER
E-	Well developed microvilli		

36-	**EM of cells of PCT showing the following except one ?**		
A-	Apex has many microvilli	B-	Basal infolding with mitochondria
C-	Apical canaliculi and pinocytotic vesicles	D-	Many fat droplets
E-	Extensive lateral membrane interdigitions		

37-	**As regard loops of Henle, only one is wrong?**
A-	It is part of the nephron
B-	Extends from PCT to join the DCT
C-	Juxtaglomerular nephron very short
D-	Its inferior U shape part is lined by simple squamous epithelium
E-	Concentrate urine by countercurrent mechanism

38-	**Structure that extends from PCT to join DCT is?**		
A-	Ducts of Billini	B-	Loop of Henle
C-	Minor calyx	D-	Loop of Bertin
E-	Juxtaglomerular complex		

39-	**As regard loops of Henle, only one is true?**		
A-	It is highly convoluted structure	B-	Extends from collecting tubules to minor calyx
C-	Lined by simple columnar epithelium	D-	Two types according to its length
E-	Secrete rennin		

40-	**The main function of Henle's loop is?**
A-	Concentration of urine by countercurrent mechanism
B-	Their lining epithelium act as osmoreceptors
C-	Their lining epithelium act as mechanoreceptors
D-	Secretion of erythropoietin
E-	Share in formation of juxtaglomerular apparatus

إذا لم يكن لله فعلك خالصًا *** فكل بناءٍ قد بنيت خراب

فللعمل الإخلاص شرطٌ إذا أتي *** وقد وافقته سنةٌ وكتاب

41-	**As regard DCT only one is true?**
A-	More convoluted than proximal tubules
B-	Starts from descending Henle's loop
C-	End in the ducts of Billini
D-	Lined by simple columnar cells
E-	Have basal acidophilic striations and indistinct apical brush border

42-	**The principal target tissue for aldosterone is?**		
A-	Descending thin loop of Henle	B-	Ascending thin loop of Henle
C-	Distal convoluted tubule	D-	Proximal convoluted tubule
E-	Straight collecting tubules		

43-	**Macula densa in juxtaglomerular apparatus is?**		
A-	Proximal convoluted tubules	B-	Distal convoluted tubules
C-	Collecting tubules	D-	Visceral layer of Bowman capsule
E-	Parietal layer of Bowman capsule		

44-	**In the kidney, rennin is produced by which of the following?**		
A-	Cells of the PCT	B-	Lacis cells
C-	Juxtaglomerular cells	D-	Podocytes
E-	Cells of the distal convoluted tubules		

45-	**They are small supporting cells, present between the afferent and efferent arterioles and the macula densa'" identifies the cells?**		
A-	Dome cells	B-	Bertin cells
C-	Lacis cells	D-	Podocytes
E-	Intraglomerular mesangial cells		

46-	**One of the following about macula densa of Juxtaglomerular apparatus is wrong?**
A-	Their cells become tall columnar instead of cubical
B-	Its basement membrane is thin or even absence
C-	Cells become crowded with deeply stained nuclei
D-	Golgi is infranuclear with ovoid mitochondria
E-	The cells have well developed basal infolding

47-	**The function of macula densa of juxtaglomerular complex is?**		
A-	Supporting structure	B-	Phagocytosis
C-	Act as osmoreceptors	D-	Secrete rennin
E-	Secrete medullipin I		

48-	**Juxta glomerular cells are?**
A-	Located in macula densa
B-	Modified neuron
C-	Modified endothelial cells of afferent arteriole
D-	Secrete angiotensin
E-	Modified smooth muscle fibers of afferent arteriole

49-	**One of the following about juxtaglomerular apparatus is wrong?**
A-	Present at urinary pole renal corpuscles
B-	Present in some corpuscles
C-	Help in control blood pressure and electrolyte balance
D-	Secrete rennin
E-	It is formed of Lacis, macula densa and Juxtaglomerular cells

50-	**Modified Mesangeal in the juxtaglomerular apparatus are called?**		
A-	Podocytes	B-	Lacis cells
C-	Macula densa cells	D-	Juxtaglomerular cells
E-	Bertin's cells		

51-	**As regard (Lacis) cells, the following statements are true except one?**		
A-	They are extra glomerular Mesangeal cells	B-	Present besides the macula densa
C-	Modified smooth muscle fibers	D-	Have well developed cytoskeleton
E-	Supporting and phagocytic cells		

52-	**As regard cortical collecting tubules, one only is wrong?**		
A-	They are called medullary rays	B-	Lined by cubical epithelium
C-	Can absorb water under control of ADH	D-	Their cells are principle and intercalated cells
E-	Open at the renal papillae into the minor calyx		

53-	**Only one is true about cortical collecting tubules:?**		
A-	Open in minor calyx	B-	Formed by union of multiple medullary tubules
C-	Called medullary rays	D-	Lined by columnar cells (principle cells only)
E-	Continuous with loop of Henle		

﴿مَن كَانَ يُرِيدُ حَرْثَ الْآخِرَةِ نَزِدْ لَهُ فِي حَرْثِهِ وَمَن كَانَ يُرِيدُ حَرْثَ الدُّنْيَا نُؤْتِهِ مِنْهَا وَمَا لَهُ فِي الْآخِرَةِ مِن نَّصِيبٍ﴾

54-	**Only one about cortical collecting tubules is true?**
A-	Formed by union of multiple medullary tubules
B-	Present at the apex of renal pyramids
C-	Their diameter about 200 um
D-	Open at area cribrosa into the minor calyx
E-	Lined by cubical cells (principle and intercalated cells)

55-	**Modified cells of DCT in the juxtaglomerular apparatus are called?**		
A-	Podocytes	B-	Lacis cells
C-	Polkison's cells	D-	Macula densa cells
E-	Juxtaglomerular cells		

56-	**"They are union of multiple medullary tubules at the apex of renal pyramids, open at the renal papillae into the minor calyx" identifies the structure?**		
A-	Ducts of Billini	B-	Ducts of Bertin
C-	DCT	D-	Loop of Henle's
E-	Major calyx		

57-	**One of the following about Billini ducts is wrong?**
A-	Formed by union of multiple medullary tubules
B-	Present at the apex of renal pyramids
C-	Their diameter about 200 um
D-	Open at area cribrosa into the minor calyx
E-	Lined by cubical cells (principle and intercalated cells)

58-	**Only one of the following about ducts of Bellini is true?**		
A-	Formed by union of multiple nephrons	B-	Present in the cortex of the kidney
C-	Lined by transitional epithelium	D-	Constitutes the renal pyramids
E-	Open at the renal papillae into the minor calyx		

59-	**Ducts of Billini of the kidney terminate in?**		
A-	Minor calyx	B-	Major calyx
C-	Ureter	D-	Pelvis of the kidney
E-	Descending limb of Henle		

60-	**Arteriole recta of the kidney supply?**		
A-	The vas deferens	B-	The cortex of the kidney
C-	The capsule	D-	Erectile tissue of the penis
E-	The medulla of the kidney		

61-	**The efferent arterioles of the renal corpuscles are considered portal vessels Because they connect the glomerular capillaries with?**		
A-	Vasa vasorum	B-	Peritubular capillaries
C-	Vasa recta	D-	Stellate veins
E-	Penicillar arteries		

62-	**Arrange the following vessels in a proper sequence for blood flow into the kidney:** **1= renal artery. 2 arcuate artery 3 interlobar artery** **4 afferent arteriole 5 interlobular artery. 6 efferent arteriole?**		
A-	1-3-2-5-4-6	B-	1-2-3-4-5-6
C-	1-4-3-2-5-6	D-	1-5-4-3-2-6
E-	6-1-3-2-5-4		

63-	**Peritubular capillaries' are branches of?**		
A-	Arcuate arteries	B-	Interlobar arteries
C-	Interlobular arteries	D-	Afferent arterioles of the corpuscles
E-	Efferent arterioles of the corpuscles		

64-	**Arcuate arteries in the kidney are originated from?**		
A-	Capsular arteries	B-	Interlobar arteries
C-	Interlobular arteries	D-	Afferent arterioles of the corpuscles
E-	Efferent arterioles of the corpuscles		

65-	**Afferent arterioles that supply the renal corpuscles are originated from?**		
A-	Segmental arteries	B-	Interlobar arteries
C-	Capsular arteries	D-	Interlobular arteries
E-	Abdominal aorta		

66-	**Erythropoietin is secreted by the following cells?**
A-	Juxta glomerular cells
B-	Cells of macula densa
C-	Extra glomerular Mesangeal cells
D-	Intraglomerular Mesangeal cells

67-	**The ureter is lined by?**		
A-	Simple columnar epithelium	B-	Stratified cubical epithelium
C-	Transitional epithelium	D-	Simple cubical epithelium
E-	Pseudo stratified columnar with steriocilia		

68-	**One of the following about the ureter is false?**
A-	Muscular tube about 25 cm length
B-	Lined by transitional epithelium
C-	Continuation of renal pelvis
D-	Musculosa in its lower end formed of two layers
E-	Its adventia is fibro elastic C.T

69-	**Transitional epithelium lines all the following except one?**		
A-	Major calyx of the kidney	B-	Pelvis of the kidney
C-	Prostatic urethra	D-	Minor calyx of the kidney
E-	Terminal part of female urethra		

70-	**The musculosa of lower 1/4 of the ureter arranged as?**
A-	Inner circular outer longitudinal
B-	Inner longitudinal and outer circular
C-	Inner circular middle longitudinal and outer circular
D-	Longitudinal muscles only
E-	Inner longitudinal, middle circular and outer longitudinal

71-	**The musculosa of upper 3/4 of the ureter arranged as?**
A-	Inner circular outer longitudinal
B-	Inner longitudinal and outer circular
C-	Inner longitudinal, middle circular and outer longitudinal
D-	Inner circular middle longitudinal and outer circular
E-	Circular muscles only

72-	**The musculosa of urinary bladder arranged as?**
A-	Inner circular outer longitudinal
B-	Inner longitudinal, middle circular and outer longitudinal
C-	Inner longitudinal and outer circular
D-	Inner circular middle longitudinal and outer circular
E-	One layer of skeletal muscles circularly arranged

73-	**The bulbourethral (Cowper's) glands are open in?**		
A-	Prostatic urethra	B-	Membranous urethra
C-	Penile urethra	D-	At fossa naviculars
E-	Urinary Bladder		

74-	**The urinary bladder is lined by?**		
A-	Stratified columnar epithelium	B-	Stratified cubical epithelium
C-	Transitional epithelium	D-	Stratified squamous epithelium
E-	Pseudo stratified columnar with stereocilia		

75-	**Lining epithelium of prostatic urethra is?**
A-	Stratified squamous epithelium
B-	Stratified columnar epithelium
C-	Stratified columnar with patches of pseudo stratified columnar
D-	Transitional epithelium
E-	Simple columnar epithelium

76-	**The membranous urethra is lined by?**
A-	Pseudo stratified columnar epithelium with stereocilia
B-	Transitional epithelium
C-	Stratified columnar epithelium
D-	Simple cubical partially ciliated epithelium
E-	Pseudo stratified columnar epithelium with true cilia

77-	**Mesangial cells?**		
A-	share in formation of blood renal barrier	B-	line the visceral layer of Bowman's capsule
C-	secrete medullin hormone	D-	have phagocytic function

78-	**Distal convoluted tubules?**		
A-	are 3-5 cells in cross section	B-	cytoplasm is deeply acidophilic
C-	is 60 um in diameter	D-	have clear cell boundaries & no brush border

79-	**The juxtaglomerular cells are formed of?**
A-	podocytes
B-	modified smooth muscles mainly in media of afferent arteriole
C-	Intraglomerular mesangial cells
D-	columnar cells in wall of collecting tubules

80-	**The membranous urethra is lined by?**		
A-	pseudostratified columnar epithelium	B-	simple squamous epithelium
C-	stratified squamous	D-	transitional epithelium

81-	The fossa navicularis of male urethra is lined by?		
A-	pseudostratified columnar epithelium	B-	stratified columnar epithelium
C-	stratified squamous	D-	transitional epithelium

82-	The penile urethra above fossa navicularis is lined by?		
A-	pseudostratified columnar epithelium	B-	stratified columnar epithelium
C-	stratified squamous	D-	transitional epithelium

83-	The female urethra is?
A-	lined by transitional epithelium & stratified columnar epithelium
B-	a curved tube
C-	20-25 cm long
D-	pathway for semen &urine

84-	Extra glomerular mesangeal cells?
A-	contain rennin granules in cytoplasm
B-	are present between the glomerular capillaries
C-	show basal infoldings
D-	are present between afferent &efferent arterioles & macula densa cells

85-	Intra glomerular mesangeal cells?		
A-	contain rennin granules in cytoplasm	B-	show apical brush border
C-	cytoplasm is rich in lysosomes	D-	have Golgi apparatus present in basal region

86-	Medullary collecting tubules are?
A-	present in the medullary rays
B-	lined by intercalated cells only
C-	formed by the union of several cortical collecting tubules
D-	open into the tips of renal papillae

87-	A 40 years old woman complaining of urinary incontinence, this may result from lesion of?
A-	smooth muscle fibers of external urethral sphincter
B-	skeletal muscle fibers of internal urethral sphincter
C-	smooth muscle fibers of internal urethral sphincter
D-	skeletal muscle fibers of external urethral sphincter

88-	Which region of the kidney contain glomeruli?		
A-	Cortex	B-	Medulla
C-	Calyx	D-	pelvis
E-	papillea		

89-	**Regarding the structure of kidney's capsule which is TRUE?**
A-	Formed of Transitional epithelium
B-	Are dense fibrous connective tissue with smooth muscles
C-	Are loose cellular connective tissue with many lymphocytes
D-	Formed of adipose connective tissue
E-	None of the above

90-	**Stratified Squamous epithelium is present in?**		
A-	Prostatic urethra	B-	Membranous urethra
C-	Terminal part of penile urethra	D-	Initial part of penile urethra

91-	**Which of the following structures is related to Juxtaglomerular apparatus?**		
A-	Macula densa	B-	Endothelial cells of glomeruli
C-	Podocyte	D-	Loop of Henle

92-	**Regarding the lining epithelium of the Glomerular Capillaries, which of the following is true?**		
A-	fenestrated endothelium with Diaphragm	B-	Continuous endothelium
C-	fenestrated endothelium without Diaphragm	D-	Non fenestration endothelial with diaphragm

عن أبي هريرة – رضي الله عنه – أنَّ رَسولَ اللَّهِ صَلَّى اللهُ عليه وسلَّمَ، قالَ:

(الْمُؤْمِنُ القَوِيُّ خَيْرٌ وَأَحَبُّ إلى اللهِ مِنَ المُؤْمِنِ الضَّعِيفِ، وفي كُلٍّ خَيْرٌ. احْرِصْ علَى ما يَنْفَعُكَ، وَاسْتَعِنْ بالله وَلَا تَعْجِزْ، وإنْ أَصَابَكَ شَيءٌ، فلا تَقُلْ: لو أنِّي فَعَلْتُ كانَ كَذَا وَكَذَا، ولَكِنْ قُلْ: قَدَرُ اللهِ وما شَاءَ فَعَلَ؛ فإنَّ (لو) تَفْتَحُ عَمَلَ الشَّيْطَانِ)

ANSWERS – Histology									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	C	D	C	A	D	B	E	D	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	B	A	E	B	E	C	D	C	D
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
A	E	B	C	A	A	E	A	B	E
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
A	E	A	C	B	D	C	B	D	A
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
E	C	B	C	C	E	C	E	A	B
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
C	E	C	E	D	A	E	E	A	E
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**
B	A	E	B	D	A	C	D	E	E
71	**72**	**73**	**74**	**75**	**76**	**77**	**78**	**79**	**80**
B	B	C	C	D	C	D	D	B	A
81	**82**	**83**	**84**	**85**	**86**	**87**	**88**	**89**	**90**
C	B	A	D	C	C	D	A	B	C
91	**92**	**93**	**94**	**95**	**96**	**97**	**98**	**99**	**100**
A	C	-	-	-	-	-	-	-	-

Physiology

❖ PHYSIOLOGY - Function of kidney, nephrons

1-	**Renin is produced and secreted from granules located in the?**				
A-	Macula densa	B-	Mesangial cells	C-	Intercalated cells
D-	Afferent arteriole	E-	Interstitial cells		

2-	**Macula dens is present in?**						
A-	Proximal tubules	B-	Distal tubules	C-	Lope of Henle	D-	Collecting tubules

3-	**Renal blood flow is kept constant over a wide range of arterial pressures (about 60 - 180 mmHg). What is this range of pressure referred to as?**		
A-	the myogenic pressure threshold	B-	the autoregulatory range
C-	the threshold range	D-	extrinsic regulation range
E-	neurogenic regulation range		

4-	**Which two factors are controlled by tubuloglomerular feedback?**				
A-	GFR and RBF	B-	RBF and tubular load	C-	GFR and tubular load
D-	Filtration fraction and RBF	E-	GFR and filtration fraction		

5-	**Which of the following structure is related to juxtaglomerular apparatus?**		
A-	macula densa	B-	Podocyte
C-	Intra raglomerular mesangial cells	D-	loop of Henel

6-	**concerning the macula densa?**
A-	The macula densa senses the total amount of sodium chloride of tubular urine being delivered by the loop of Henle into the distal convoluted tubule
B-	Renin release causes intravascular angiotensin formation and indirectly aldosterone release
C-	If the delivery of sodium chloride is normal the macula densa signals to arteriole (pre-glomerular) to release renin
D-	all is correct

7-	**If the systemic blood pressure increased by 20 mmHg?**
A-	Glomerular capillary pressure increases by 20 mmHg
B-	GFR increases dramatically
C-	Afferent arteriole constricts
D-	Efferent arteriole constricts

"كم أماتت رغبةُ الكمال إنجازَ كثيرٍ من جليل الأعمال!

كما أمات التّراخي والتسويفُ كثيرًا من فريدِ التأليف!"

8-	**Which of the following is true about renal blood flow?**
A-	The kidneys receive 12-15% of cardiac output
B-	Is kept constant within a small range by sympathetic vasomotor tone
C-	Measured by renal clearance of inulin
D-	The dilution method is used for its measurement
E-	Goes mainly to the renal cortex

9-	**The following hormones can be formed by the kidney EXCEPT?**		
A-	Erythropoietin	B-	Angiotensin II
C-	Prostaglandins	D-	ADH
E-	Calcitrol		

10-	**Which of the following is NOT a function of the kidneys?**
A-	Regulation of extracellular volume
B-	Regulation of plasma glucose concentration
C-	Regulation of arterial blood pressure
D-	Excretion of end products of protein metabolism
E-	Excretion of foreign chemicals (eg. Pharamaceuticals, food additives, etc)

11-	**Which of the following increases the glomerular filteration rate?**
A-	An increase in Bowman's capsule pressure
B-	A decrease in glomerular capillary hydrostatic pressure
C-	An increase in tubular fluid hydrostatic pressure
D-	A decrease in efferent arteriolar resistance
E-	An increase in plasma colloid pressure

12-	**An adrenal tumor that causes excess aldosterone secretion would tend to _____ plasma K+concentration, _____ plasma pH,_____ renin secretion, and _____ blood pressure?**		
A-	Decrease, decrease, decrease, decrease	B-	Decrease, increase, decrease, increase
C-	Decrease, decrease, decrease, increase	D-	Decrease, increase, increase, increase
E-	Increase, increase, decrease, increase		
F-	Increase, decrease, decrease, increase		

13-	**Renin is produced and secreted from granules located in the?**		
A-	Macula densa	B-	Mesangial cells
C-	Intercalated cells	D-	Afferent arteriole

14-	**The kidneys are?**		
A-	Help regulate blood volume	B-	Help control blood pressure
C-	Help control PH	D-	All of the above are correct

15-	**The renal blood flow?**
A-	Is 40% of the cardiac output?
B-	Reflects the high oxygen consumption by the kidney
C-	Increases under stressfu1 conditions
D-	Is more to the cortex than to the medulla

16-	**Angiotensin II?**		
A-	Stimulates aldosterone secretion	B-	Causes vasoconstriction of efferent arterioles
C-	Stimulates thirst center	D-	All of the above

17-	**Stimulation of renin secretion will increase the?**		
A-	K concentration in the blood	B-	ECF volume
C-	Hematocrite value	D-	Plasma colloid oncotic pressure

18-	**The juxta glomerular apparatus?**
A-	Is present at the beginning of the proximal convoluted tubule
B-	Formed of the thin segment of the loop of Henle and the vasa recta
C-	Is richly supplied by cholinergic nerves
D-	None of the above

19-	**Juxtamedullary nephrons?**		
A-	Are 90% of total nephrons	B-	Have short loops of Henle
C-	Cortex lie in medulla	D-	Have important role in concentration of urine
E-	Are 40 % of total nephrons		

20-	**The macula densa cells of Juxtaglomerular apparatus?**		
A-	Are present in afferent arterioles	B-	Are present in efferent arterioles
C-	Secrete rennin	D-	Secrete aldosterone
E-	Are receptor cells that respond to change in Na CL		

21-	**Renin secretion is increased by the following EXCEPT?**		
A-	Hypovolemia	B-	Hyponatremia in DCT
C-	Sympathelic stimulation	D-	Hyponatremia in PCT

22-	**The following hormones are secreted the kidney EXCEPT?**		
A-	Renin	B-	Angiotensin I
C-	Erythropoietin	D-	1.25 DHCC

23-	**Concerning the macula densa?**
A-	The macula densa senses the total amount of sodium chloride (sodium chloride concentration times volume) of tubular urine being delivered /by the loop of Henle into the distal convoluted tubule
B-	If the delivery of sodium chloride is lower than normal the macula densa signals the afferent arteriole (pre-glomerular) to release renin
C-	Renin release causes intravascular angiotensin formation and indirectly aldosterone release
D-	All are correct

24-	**The role of the kidney in homeostasis may include which of the following?**
A-	Regulation of extracellular fluid composition
B-	Regulation of red blood cell formation
C-	Secretion of certain hormones, such as angiotension II, prostaglandins, and kinins in the regulation of blood pressure
D-	Secretion of erythropoietin
E-	All of the above

25-	**All of the following about the nephron is true, EXCEPT?**
A-	Has endocrine function
B-	Is lined by a single layer of epithelial cells throughout its length
C-	Is under endocrine control
D-	Lies exclusively within the renal cortex

26-	**The glomerulus?**
A-	Has both afferent and efferent arterioles
B-	Contains capillaries which are at a higher hydrostatic pressure
C-	Contains renin-secreting cells
D-	Is lined by fenestrated endothelium
E-	All of the above

27-	**The glomerulus?**
A-	Is permeable to negatively charged large molecules
B-	Contains no active transport systems (pumps) that produce an important effect on the composition of the glomerular filtrate
C-	Produces a filtrate with higher concentration of amino acids than found in plasma
D-	Produces a filtrate with a lower concentration of urea than found in plasma

28-	**The following is (are) a feature of the glomerulus?**		
A-	Podocyte	B-	Fenestrated epithelium
C-	Slit membrane	D-	Foot processes
E-	All of the above		

29-	**The loops of Henle of the outer cortical nephrons?**
A-	Do not play any important role in overall function, and are simply unimportant
B-	Do not participate in the urinary diluting mechanism
C-	Are functionally unimportant in the renal conservation of sodium and water
D-	Do not contribute to the medullary osmotic gradient

30-	**Experiment destruction of the juxtaglomerular cells results in loss of production of?**		
A-	Aldosterone	B-	Angiotensinogen
C-	Erythropoietin	D-	Renin

31-	**The juxtaglomerular apparatus?**
A-	Is present in the proximal convoluted tubule
B-	Secretes a converting enzyme that acts on angiotensin I forming angiotensin II
C-	Is formed of the macula densa and the juxtaglomerular cells
D-	Is stimulated by increased blood flow through the afferent arteriole

32-	**The function of the macula densa cells is to?**
A-	Prevent water reabsorption in the ascending limb of the loop of henle
B-	Add bicarbonate ions to the filtrate
C-	Secrete renin
D-	Monitor filtrate NaCl

33-	**Renal blood flow is?**		
A-	0.8 1.2 l/min	B-	1.2-1.3 L/Min
C-	1.5-2 L/Min	D-	2-2.5 L/Min

34-	**Inulin clearance closely resembles?**		
A-	GFR	B-	Renal plasma flow
C-	Creatinine clearance	D-	P.A.H clearance

35-	**Which two factors are controlled by tubuloglomerular feedback?**		
A-	GFR and RBF	B-	RBF and tubular load
C-	GFR and tubular load	D-	Filtration fraction and RBF

36-	The net filtration pressure for glomerular filtration is?		
A-	10 mm Hg	B-	25 mm Hg
C-	50 mm Hg	D-	100 mm Hg

ANSWERS – Function of kidney, nephrons									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	B	B	A	A	D	C	E	D	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
E	B	D	D	D	D	B	D	D	E
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
D	B	D	E	D	E	B	E	D	D
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
C	D	B	C	A	A	-	-	-	-

يقول الإمام الغزالي – رحمه الله – في كتابه بداية الهداية :

ولا ينبغي أن تكون اوقاتك مهملة فتشتغل في كل وقت بما اتفق كيف اتفق، بل ينبغي أن تحاسب نفسك وترتب أورادك في ليلك ونهارك، وتعين لكل وقت شغلا لا تتعداه، ولا تؤثر فيه سواه فبذلك تظهر بركة الأوقات .

فأما إذا تركت نفسك سدى مهملا إهمال البهائم لا تدري بماذا تشتغل في كل وقت، فينقضي أكثر أوقاتك ضائعا، وأوقاتك عمرك، وعمرك رأس مالك، وعليه تجارتك، وبه وصولك إلى نعيم دار الأبد في جوار الله تعالى؛ فكل نفس من أنفاسك جوهرة لا قيمة لها؛ إذ لا بدل له فإذا فات فلا عودة له؛ فلا تكن كالحمقى المغرورين الذين يفرحون كل يوم بزيادة أموالهم مع نقصان أعمارهم، فأي خير في مال يزيد وعمر ينقص ! ولا تفرح إلا بزيادة علم أو عمل صالح فإنهما رفيقاك بصحبانك في القبر حيث يتخلف عنك أهلك ومالك، وولدك، وأصدقاؤك .

❖ Urine formation, GFR

1-	Renal blood flow is kept constant by autoregulation in the mean systemic blood pressure range of?						
A-	0-90 mmHg	B-	60-180 mmHg	C-	200-250 mmHg	D-	All of the above

2-	Normal glomerular filtration rate is?						
A-	25 ml/min	B-	50 mllmin	C-	75 mllmin	D-	125 mllmin

3-	Glomerular filtration rate would be increased by?
A-	constriction of the afferent arteriole
B-	a decrease in afferent arteriolar pressure
C-	compression of the renal capsule
D-	a decrease in the concentration of plasma protein

4-	Which of the following is (are) responsible for the high (GFR)?
A-	Low blood hydrostatic pressures in the glomerular capillaries
B-	High permeability of the glomerular capillaries
C-	High hydrostatic pressure of the fluid in the Bowman's capsule
D-	The renin-angiotensin-aldosterone system
E-	High rate of water reabsorption across the walls of the proximal tubules

5-	Which of the following changes tends to increase glomerular filtration rate (GFR)?
A-	Increased afferent arteriolar resistance
B-	Decreased efferent arteriolar resistance
C-	Increased glomerular capillary filtration coefficient
D-	Increased Bowman's capsule hydrostatic pressure

6-	All of the following are criteria of a substance used for GFR measurement, EXCEPT?		
A-	Should not be secreted by the tubular cells	B-	Should be secreted by the tubular cells
C-	Should not be reabsorbed by the tubular cells	D-	Should not be toxic

7-	Which substance is filtered most easily by the glomerular capillaries?		
A-	Albumin in plasma	B-	Negatively charged molecules
C-	Molecules with a diameter less than 4 nm	D-	Red blood cells

8-	Which of the following has a rate of urinary excretion (units of solute per unit time) that is almost always identical to its rate of glomerular filtration in a healthy adult?						
A-	Sodium	B-	Glucose	C-	chloride	D-	Creatinine

9-	Which of the following causes decrease in GFR?		
A-	Increase plasma colloid pressure	B-	Increase plasma hydrostatic pressure
C-	Decrease afferent resistance	D-	Decrease pressure inside bowman's capsule

10-	In a healthy adult man the glomerular filteration rate is expected to be?		
A-	225 ml/min	B-	325 ml/min
C-	125 ml/min	D-	25 ml/min
E-	180 ml/min		

11-	The following factors when increased, they will increase the glomerular filteration rate EXCEPT?		
A-	Renal blood flow	B-	Afferent arteriolar diameter
C-	Efferent arteriolar diameter	D-	Permeability of glomerular membrane
E-	Capillary surface area		

12-	Renal blood flow is?
A-	Increased when renal sympathetic nerves are stimulated
B-	Autoregulated when mean arterial blood pressure is between 60-180 mmHg
C-	Measured by urea clearance
D-	About 2 liters in a normal man
E-	Decreased when the fluid flow at the site of the macula densa is decreased

13-	The following decreases the GFR?
A-	Increased constriction of the afferent arteriole
B-	Decrease in the pressure of the afferent arteriole
C-	Decrease in concentration of plasma protein
D-	Decreased hydrostatic pressure in Bowman's capsule
E-	Decrease in the radius of efferent arterioles

14-	Use the values below to answer the following question. Glomerular capillary hydrostatic pressure = 47 mm Hg Bowman space hydrostatic pressure = 10 mm Hg Bowman space oncotic pressure = 0 mm Hg At what value of glomerular capillary oncotic pressure would glomerular filtration stop?		
A-	57 mm Hg	B-	47 mm Hg
C-	37 mm Hg	D-	10 mm Hg
E-	0 mm Hg		

15-	**Which of the following would cause an increase in both glomerular filtration rate (GFR) and renal plasma flow (RPF)?**		
A-	Hyperproteinemia	B-	A ureteral stone
C-	Dilation of the afferent arteriole	D-	Dilation of the efferent arteriole
E-	Constriction of the efferent arteriole		

16-	**Which of the following would cause the greatest decrease in GFR in a person with otherwise normal kidneys?**
A-	Decrease in renal arterial pressure from 100 to 80 mm Hg in a normal kidney
B-	50% increase in glomerular capillary filtration coefficient
C-	50% increase in proximal tubular sodium reabsorption
D-	50% decrease in afferent arteriolar resistance
E-	50% decrease in efferent arteriolar resistance
F-	5 mm Hg decrease in Bowman's capsule pressure

17-	**Which change tends to increase GFR?**
A-	Increased afferent arteriolar resistance
B-	Decreased efferent arteriolar resistance
C-	Increased glomerular capillary filtration coefficient
D-	Increased Bowman's capsule hydrostatic pressure
E-	Decreased glomerular capillary hydrostatic pressure

18-	**The GFR of a 26-year-old man with glomerulonephritis decreases by 50% and remains at that level for one month. For which substance would you expect to find the greatest increase in plasma concentration?**		
A-	Creatinine	B-	K+
C-	Glucose	D-	Na+
E-	Phosphate		
F-	H+		

19-	**Which changes would you expect to find after administering a vasodilator drug that caused a 50% decrease in afferent arteriolar resistance and no change in arterial pressure?**
A-	Decreased renal blood flow, decreased GFR, and decreased peritubular capillary hydrostatic pressure
B-	Decreased renal blood flow, decreased GFR, and increased peritubular capillary hydrostatic pressure
C-	Increased renal blood flow, increased GFR, and increased peritubular capillary hydrostatic pressure
D-	Increased renal blood flow, increased GFR, and no change in peritubular capillary hydrostatic pressure
E-	Increased renal blood flow, increased GFR, and decreased peritubular capillary hydrostatic pressure

20-	**If the average hydrostatic pressure in the glomerular capillaries is 50 mm Hg, the hydrostatic pressure in the Bowman's space is 12 mm Hg, the average colloid osmotic pressure in the glomerular capillaries is 30 mm Hg, and there is no protein in the glomerular ultrafiltrate,what is the net pressure driving glomerular filtration?**		
A-	8 mm Hg	B-	32 mm Hg
C-	48 mm Hg	D-	60 mm Hg
E-	92 mm Hg		

21-	**Which change tends to increase peritubular capillary fluid reabsorption?**		
A-	Increased blood pressure	B-	Decreased filtration fraction
C-	Increased efferent arteriolar resistance	D-	Decreased angiotensin II
E-	Increased renal blood flow		

22-	**Which change, compared with normal, would be expected to occur,under steady-state conditions, in a patient whose severe renal disease has reduced the number of functional nephrons to 25% of normal?**
A-	Increased GFR of the surviving nephrons
B-	Decreased urinary creatinine excretion rate
C-	Decreased urine flow rate in the surviving nephrons
D-	Decreased urinary excretion of sodium
E-	Increased urine-concentrating ability

23-	**A selective decrease in efferent arteriolar resistance would _____glomerular hydrostatic pressure, _____ GFR, and _____renal blood flow.**		
A-	Increase, increase, increase	B-	Increase, decrease, increase
C-	Increase, decrease, decrease	D-	Decrease, increase, decrease
E-	Decrease, decrease, increase	F-	Decrease, increase, increase

24-	**Increases in both renal blood flow and GFR are caused by which mechanism?**		
A-	Dilation of the afferent arterioles	B-	Increased glomerular capillary filtration coefficient
C-	Increased plasma colloid osmotic pressure	D-	Dilation of the efferent arterioles

25-	**Renal blood flow is kept constant by autoregulation in the mean systemic blood pressure range of?**		
A-	0-90 mmHg	B-	60-180 mmHg
C-	200-250 mmHg	D-	All of the above

26-	**Glomerular filtration rate would be increased by?**		
A-	Constriction of the afferent arteriole	B-	A decrease in afferent arteriolar pressure
C-	Compression of the renal capsule	D-	A decrease in the concentration of plasma protein

27-	**Which of the following is responsible for the high (GER)?**
A-	Low blood hydrostatic pressures in the glomerular capillaries
B-	High permeability of the glomerular capillaries
C-	High hydrostatic pressure of the fluid in the Bowman's capsule
D-	High rate of water reabsorption across the walls of the proximal tubules

28-	**Renal blood flow?**
A-	Glomerular capillaries drain into peritubular veins
B-	Renal autoregulation is prevented by denervation
C-	Kidney's receive approximately 15% CO
D-	Angiotensin II causes greater constriction of the efferent than the afferent arteriole

29-	**Renal blood flow is kept constant over a wide range of arterial pressures (about 60-180 mmHg). What is this range of pressure referred to as?**		
A-	The myogenic pressure threshold	B-	The autoregulatory range
C-	The threshold range	D-	Extrinsic regulation range

30-	**The permeability of the glomerular capillary is?**
A-	Less than the permeability of skeletal Imuscles capillaries
B-	10-40 times more than the permeability .skeletal muscles
C-	50 Times more than permeability of skeletal muscle
D-	1000-4000 times more than the permeabilityof skeletal muscles

31-	**The glomerular membrane is impermeable for plasma proteins due to?**		
A-	Electrical charges	B-	Molecular weight
C-	Both a & b	D-	Proteins-are-water insoluble

32-	**About the glomerular filtration rate, all of the following are correct EXCEPT?**
A-	It is about 125 ml 7 minute
B-	It depends mainly on the glomerular capillary hydrostatic pressure
C-	It is increased by vasodilatation of afferent arterioles
D-	It is increased by increased plasma proteins Oncotic pressure

33-	**All of the following about the glomerular capillary oncotic pressure is correct EXCEPT?**
A-	It is the osmotic pressure of plasma proteins
B-	It is about 25 mm Hg
C-	Its increase increases the GFR
D-	It increases along the length of the glomerular capillaries

34-	**The glomerular filtration rate?**
A-	Increases with decreased filtration coefficient
B-	Increases with increased sympathetic tone to the afferent arteriole
C-	Decreases gradually along the length of the glomerular membrane
D-	Increases during obstruction of the ureter by a stone

35-	**Which of the following is correct about glomerular capillary hydrostatic pressure?**
A-	Is increased with afferent arteriolar vasoconstriction
B-	Is on the average 25 mmHg
C-	Ls the force, which opposes filtration?
D-	Rises with constriction of the efferent arteriole

36-	**The rate of excretion of any substance by the kidney equals?**
A-	The rate of filtration
B-	The rate of filtration – rate of secretion
C-	The rate of filtration rate of absorption rate of secretion
D-	Rate of absorption + rate of secretion

37-	**Sympathetic stimulation causes all of the following EXCEPT?**		
A-	Increase in GFR	B-	Increase in rennin secretion
C-	Increase afferent arteriolar resistance	D-	Increased renal sodium reabsorption

38-	**GFR is decreased by?**		
A-	Increase renal blood flow	B-	Dilatation of afferent arteriole
C-	Sympathetic stimulation	D-	Increase ABP
E-	Increase filtration coefficient		

39-	**The renal blood flow is?**
A-	5% of the cardiac output
B-	Higher in the medulla than in the cortex
C-	Markedly affected by artrial blood pressure whatever its level
D-	Decrease by sympathetic stimulation

40-	**Factor affecting GFR are?**
A-	Changes in renal blood flow
B-	Changes in glomerular capillary hydrostatic pressure
C-	Ureteric obstruction
D-	All of the above

41-	**The following factor(s) tend(s) to INCREASE the rate of glomerular filtration (GFR)?**
A-	Decreased albumin concentration in plasma
B-	Vasodilation of the afferent (pre-glomerular) arteriole
C-	Vasoconstriction of the efferent (post-glomerular) arteriole
D-	All are correct

42-	**The following factor(s) tend(s) to INCREASE the rate of glomerular filtration (GFR)?**
A-	Sympathetic stimulation (norepinephrine) of the afferent arteriole
B-	Obstruction of the renal tubules, ureter or urethra
C-	Vasodilation of the efferent arteriole
D-	None are correct

43-	**With respect to glomerular filtration?**
A-	Formed by passive process of ultra-filtration through pores glomerular membrane
B-	Not affected by RBF
C-	It is about 127 L/min
D-	All of the above

44-	**If the net transport rate = filtration rate - excretion rate; and if the transport rate is positive?**		
A-	Secretion must have taken place	B-	Reabsorption must have taken place
C-	Excretion rate is in excess of filtration rate	D-	None of the above

45-	**Glomerular filtration rate increases by?**		
A-	Afferent arteriolar constriction	B-	Mild Efferent arteriolar constriction
C-	Reducing renal blood flow	D-	Sympathetic stimulation

46-	**All about glomerular filtration is true, EXCEPT?**
A-	Normally occurs at an overall rate of about 120 mllmin
B-	Pressure gradient equals the difference between the hydrostatic and plasma oncotic pressures in the glomerular capillary
C-	Fluid (ultrafiltrate) has the same composition as plasma
D-	Increases the hematocrit in glomerular capillary blood

47-	**Solute particles move from the plasma of renal glomerulus to the fluid in the bowman's capsule by?**		
A-	Bulk flow	B-	Active transport
C-	Diffusion	D-	Renal flow
E-	A and C are correct		

48-	Glomerular filtration rate would be decreased by all of the following, EXCEPT?		
A-	Chronic hypertension	B-	Hypoproteinemia
C-	Constriction of the afferent arteriole	D-	Obstruction of a ureter

49-	Under normal conditions all the following statements are correct, EXCEPT?
A-	The colloidal osmotic pressure of plasma in the efferent arteriole is higher than in the afferent arteriole
B-	The pressure in Bowman's capsule equals approximately 10 mmHg
C-	The net filtration pressure equals approximately 10 mmHg
D-	An increase in the afferent arteriolar tone increases the net glomerular filtration pressure

50-	The initial step urine formation is the ultrafiltration at the glomeruli, the forces aiding ultrafiltration include?		
A-	Colloid osmotic pressure of plasma protein	B-	Glomerular capillary pressure
C-	Hydrostatic pressure in bowman's capsule	D-	Crystalloid osmotic pressure of the final urine

51-	In a normal kidney, a large increase in glomerular filtration rate would be expected to occur following?
A-	An increase in mean arterial pressure from 90 mmHg to 140 mmHg
B-	Vasoconstriction of glomerular afferent arterioles
C-	Strong, acute sympathetic stimulation to kidney
D-	Substantial increases in renal blood flow

52-	Glomerular filtration rate (GFR) and renal blood flow (RBF) will both be increased if?
A-	The efferent arid afferent arterioles are both dilated
B-	The efferent and afferent arterioles are both constricted
C-	Only the afferent arteriole is constricted
D-	Only the efferent arteriole is constricted

53-	Regarding the glomerular filtration rate (GFR)?
A-	The GFR depends on the pressure in the afferent arteriole
B-	A substance that has a clearance less than the GFR must have been secreted by the renal tubules
C-	The GFR can be determined by measuring the clearance of insulin
D-	The glomerular filtrate has the same composition as plasma

54-	Regarding the renal circulation?
A-	The kidneys receive about 10% of the resting cardiac output
B-	As cardiac output increases during exercise renal blood flow rises
C-	Renal blood flow is maintained within narrow limits despite changes in ABP
D-	Increased Sympathetic activity results in increased blood flow to the kidneys

55-	**There is very little protein in the glomerular filtrate because?**
A-	All serum proteins are too large to fit through the glomerular pores
B-	Of a combination of pore size and negative charges lining the pores
C-	Of active reabsorption of filtered protein by the glomerular epithelial cells
D-	Of metabolism of filtered proteins by the urinary flora

56-	**The clearance rate for a substance that is freely filtered, but neither secreted nor reabsorbed by the kidney is equal to the?**		
A-	Filtration fraction	B-	Renal plasma flow
C-	Glomerular filtration rate	D-	Urinary excretion rate of the substance

57-	**If the diameter of the afferent arteriole << the diameter of the efferent arteriole, which of the following will result?**		
A-	Decreased Net flow pressure	B-	Increased glomerular pressure
C-	Increased GFR	D-	Increased [Na] in the filtrate reaching the PCT

58-	**Which of the following changes would you expect to find in a person deprived of water for 24 hours?**
A-	Decreased plasma renin activity
B-	Decreased plasma antidiuretic hormone concentration
C-	Increased plasma atrial natriuretic peptide concentration
D-	Increased water permeability in the collecting duct

ANSWERS – Urine formation, GFR									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	D	D	B	C	B	C	D	A	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	B	A	C	C	E	C	A	D	A
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
C	A	E	A	B	D	B	D	B	C
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
C	D	C	C	D	C	A	C	D	D
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
D	D	A	B	B	B	E	B	D	B
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
D	D	A	C	B	C	A	D	-	-

❖ Tubular reabsorption

1-	**The characteristics of the Loop of Henle include?**
A-	Descending limb is not permeable to water
B-	Urine becomes more dilute as it reaches the tip of the loop
C-	Ascending limb is permeable to water
D-	In thin ascending limb urea is actively transported to interstitial fluid
E-	Na is reabsorbed by Na-K ATPase in ascending limb

2-	**About Bicarbonate Reabsorption?**
A-	Represents a high capacity low gradient system
B-	Filtered bicarbonate/ day= 400 – 500 mmol/day
C-	More HCO3 is absorbed in distal than proximal tubules
D-	PCT apical membrane is permeable to HCO
E-	H2CO3 is PCT lumen is catalyzed at brush border by carbonic anhydrase to give CO2 and HCO3-

3-	**The functions of the renal proximal convoluted tubes include?**
A-	Reabsorption of all bicarbonate filtered at the glomerulus
B-	Production of renin
C-	Reaborption of 80% of filtered bicarbonate
D-	Reabsorption of bicarbonate resulting in fixed acid secretion
E-	Hydrogen ion secretion by facilitated diffusion

4-	**The tonicity of urine as it enters the renal collecting duct is?**		
A-	Isotonic	B-	Hypotonic or isotonic, but never hypertonic
C-	Hypotonic	D-	Hypertonic
E-	Hypertonic or isotonic, but never hypotonic		

5-	**Sodium reabsorption in the proximal tubules from the luminal surface is marked by all of the following EXCEPT?**		
A-	Sodium glucose co-transporters	B-	Sodium-hydrogen exchanger
C-	Na-K pump	D-	Solvent drag
E-	Sodium amino acid transporters		

6-	**Glomerular filtration rate would be increased by?**
A-	A decrease in afferent arteriolar pressure
B-	A decrease in renal blood flow
C-	A decrease in efferent arteriolar resistance
D-	Decrease in the concentration of plasma proteins
E-	Obstruction of ureters

7-	**Potassium secretion in the distal tubule will be increased if there is an increase in?**		
A-	Plasma (H+)	B-	Plasma volume
C-	Plasma sodium concentration	D-	Plasma potassium concentration
E-	Plasma calcium concentration		

8-	**In the proximal tubules?**
A-	20% of filtered amino acid is reabsorbed
B-	Sodium reabsorption is aldosterone dependent
C-	Equal percentage of sodium and water is reabsorbed
D-	Potassium is mainly secreted
E-	Calcium cannot be reabsorbed

9-	**In normal kidneys, which of the following is true of the osmolarity of renal tubular fluid that flows through the early distal tubule in the region of the macula densa?**		
A-	Usually isotonic compared with plasma	B-	Usually hypotonic compared with plasma
C-	Usually hypertonic compared with plasma	D-	Hypertonic, compared with plasma, in antidiuresis

10-	**Select the correct answer about proximal tubules?**
A-	K+ is secreted in exchange with the Na+ which is reabsorbed under the effect of aldosterone
B-	Glucose, amino acids & proteins are completely reabsorbed
C-	Only 10% of the filtered water is reabsorbed
D-	Parathormone increase phosphate reabsorption

11-	**When a person is dehydrated, hypotonic fluid will be found in the?**		
A-	Glomerular filtrate	B-	Proximal tubule
C-	Descending lobe of loop of Henle	D-	Distal convoluted tubule

12-	**Sodium reabsorption is coupled with the following substance in proximal convoluted tubule?**		
A-	Hydrogen ion	B-	Amino acid
C-	Glucose	D-	All of the above

13-	**Glucose reabsorption?**
A-	Occurs in every segment of the renal tubules
B-	Is passive and results from potassium reabsorption
C-	Is coupled to sodium reabsorption
D-	Is complete when the plasma glucose level is 250 mg%

14-	**Glucose reabsorption in the renal tubules?**
A-	Has no Tm
B-	Across the luminal border of the proximal convoluted tubule is by facilitated diffusion
C-	Is coupled to sodium reabsorption in the proximal convoluted tubules
D-	Occurs equally in every segment of the nephron

15-	**In the proximal convoluted tubules?**
A-	Only 30% of the filtered water is reabsorbed
B-	The luminal fluid is iso-osmotic to plasma
C-	Hydrogen ions are actively reabsorbed
D-	Potassium ions secretion is controlled by aldosterone

16-	**About the proximal convoluted tubule?**
A-	It absorbs about 50% of the glomerular filtrate
B-	Absorbs all filtered glucose and amino acids
C-	At the end of the tubule, the tubular fluid becomes hyperosmotic
D-	All of the above

17-	**Renal handling of Na+ in the PCT includes all of the following EXCEPT?**
A-	About 70% of filtered load is reabsorbed
B-	Na+ reabsorption at the luminal border is an active process
C-	Na+ reabsorption is accompanied by glucose and amino acid co-transport
D-	In the late proximal tubule Na reabsorption is accompanied by Cl-reabsorption

18-	**Under normal conditions, all of the following substances could be detected in urine EXCEPT?**		
A-	Na	B-	HC03-
C-	Amino acids	D-	Urea

19-	**Glucose reabsorption?**		
A-	Occurs in DCT	B-	Occurs in the Loop of Henle
C-	Occurs in PCT	D-	Is passive at the luminal border
E-	Is active at the basolateral border		

20-	**Glucose is transported through?**		
A-	Na anteport	B-	Na Co-transport
C-	K anteport	D-	K symport

21-	**Most water reabsorption in nephron occurs in the?**		
A-	PCT	B-	Collecting duct
C-	Loop of Henle	D-	DCT

22-	**Bicarbonate is maximally absorbed in?**		
A-	PCT	B-	DCT
C-	Collecting duct	D-	Thick ascending loop of Henle

23-	**Concerning water re-absorption?**
A-	In PCT is under the control of ADH
B-	Its obligatory type is called iso-osmotic water re-absorption
C-	Not related to solute re-absorption
D-	None of the above

24-	**The proximal convoluted tubules?**
A-	Reabsorb 30% of the water and salts of the glomerular filtrate
B-	Reabsorb all the glucose in the glomerular filtrate
C-	Contain juxtaglomerular cells which secrete rennin
D-	Are the main target cells for antidiuretic hormone (ADH)

25-	**Which statement is true concerning the thin descending limb of loop Henle?**
A-	This segment is found only in the deep cortical (iuxtarnedullary) nephrons
B-	Passive water absorption occurs in this segment
C-	Active sodium and chloride reabsorption occurs in this segment
D-	Tubular fluid osmolarity decreases from the proximal to distal portions of this segment

26-	**Which of the following is most likely to be seen if a sample of fluid leaving The proximal tubule in a person with normal kidneys is analyzed?**		
A-	It will be hypertonic compared with plasma	B-	It will have more K+ than plasma
C-	It will have more HC03- ions than plasma	D-	It will have no amino acids

27-	**Diuretics, such as acetazolamide, which produce their effect by inhibiting carbonic anhydrase, inhibit the reabsorption of sodium in?**		
A-	The proximal tubule	B-	The thick ascending limb of Henle's loop
C-	The distal convoluted tubule	D-	The cortical collecting duct

28-	**Which of the following is the primary site of salt and water reabsorption in the kidney?**		
A-	Collecting duct	B-	Glomerulus
C-	Juxtaglomerular apparatus	D-	Proximal tubule

29-	**Which of the following renal sites is characterized by low water permeability?**		
A-	Collecting duct	B-	Glomerulus
C-	Thick ascending limb of loop of Henle	D-	Proximal tubule

30-	In which one of the following situations is urinary flow less than normal?		
A-	Diabetes insipidus	B-	Diabetes mellitus
C-	Sympathetic stimulation	D-	Increased renal arterial pressure

ANSWERS – Tubular reabsorption									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
E	E	C	B	C	D	D	C	B	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	D	C	C	B	B	B	C	C	B
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
A	A	B	B	B	D	A	D	C	C

مِنْ كُلِّ شَيْءٍ إِذَا ضَيَّعْتَهُ عِوَضٌ . . . وَمَا مِنَ اللَّهِ إِنْ ضَيَّعْتَ مِنْ عِوَضِ

فَاللَّهُ سُبْحَانَهُ يُعَوِّضُ عَنْ كُلِّ شَيْءٍ مَا سِوَاهُ وَلَا يُعَوِّضُ مِنْهُ شَيْءٌ، وَيُغْنِي عَنْ كُلِّ شَيْءٍ وَلَا يُغْنِي عَنْهُ شَيْءٌ، وَيُجِيرُ مِنْ كُلِّ شَيْءٍ وَلَا يُجِيرُ مِنْهُ شَيْءٌ، وَيَمْنَعُ مِنْ كُلِّ شَيْءٍ وَلَا يَمْنَعُ مِنْهُ شَيْءٌ، كَيْفَ يَسْتَغْنِي الْعَبْدُ عَنْ طَاعَةِ مَنْ هَذَا شَأْنُهُ طَرْفَةَ عَيْنٍ ؟ وَكَيْفَ يَنْسَى ذِكْرَهُ وَيُضَيِّعُ أَمْرَهُ حَتَّى يُنْسِيَهُ نَفْسَهُ، فَيَخْسَرُهَا وَيَظْلِمُهَا أَعْظَمَ الظُّلْمِ ؟ فَمَا ظَلَمَ الْعَبْدُ رَبَّهُ وَلَكِنْ ظَلَمَ نَفْسَهُ، وَمَا ظَلَمَهُ رَبُّهُ وَلَكِنْ هُوَ الَّذِي ظَلَمَ نَفْسَهُ .

❖ Regulation of tubular reabsorption, renal handling of important substance

1-	**25 year old male running, become very dehydrated with very high secretion of ADH, the maximum water reabsorption is in?**		
A-	Cortical collecting tubule	B-	PCT
C-	DCT	D-	Loop of henle

2-	**Secretion of H+ ions in the PCT is associated with:**		
A-	Bicarbonate reabsorption	B-	Reabsorption of K
C-	Decrease PH below 4.4	D-	

3-	**The following is true about glucose reabsorption in the renal tubules?**
A-	Transport maximum is 80 mg/min
B-	The transport maximum is dependent on the amount of glucose filtered
C-	It's mainly by symport with k
D-	It's reduced with metabolic inhibitors, that reduce the activity of Na/K Atpase pump

4-	**When Glucose level is more than Tmg, this means?**
A-	Glucose excreted is equal to Glucose filtered
B-	Glucose lever in artery equal Glucose level in vein
C-	Excretion of Glucose Increases with Increasing its plasma concentration

5-	**which of the following is true?**
A-	reabsorption of water is normally less in descending limb of loop of Henle than in Ascending loop
B-	reabsorption of sodium is normally less in ascending limb of loop of Henle than in descending loop
C-	Sodium is absorbed in all segments of Nephron except in thin descending loop of Henle

6-	**Movement of urea in collecting duct is facilitated by?**				
A-	Noradrenaline	B-	Dopamine	C-	Vasopressin
D-	Angiotensin II	E-	All of the above		

7-	**Select the correct answer about proximal tubules?**
A-	K+ is secreted in exchange with the Na+ which is reabsorbed under the effect of aldosterone
B-	glucose, amino acids & proteins are completely reabsorbed
C-	only 10% of the filtered water is reabsorbed
D-	parathormone increase phosphate reabsorption

8-	**When a person is dehydrated, hypotonic fluid will be found in the?**		
A-	glomerular filtrate	B-	proximal tubule
C-	Descending lobe of loop of Henle	D-	distal convoluted tubule

9-	**The effect of antidiuretic hormone (ADH) on the kidney is to?**
A-	increase the permeability of the distal nephron to water
B-	increase the excretion of Na
C-	increase the excretion of water
D-	increase the diameter of the renal artery

10-	**An increase in the osmolarity of the extracellular compartment will?**
A-	stimulate the volume and osmoreceptors, and inhibit ADH secretion
B-	inhibit the volume and osmoreceptors, and stimulate ADH secretion
C-	inhibit the volume and osmoreceptors, and inhibit ADH secretion
D-	stimulate the volume and osmoreceptors, and stimulate ADH secretion

11-	**K+ excretion is markedly influenced by?**		
A-	aldosterone	B-	rate of tubular secretion of H
C-	amount of Na+ delivered to tubules	D-	all of the above

12-	**In the distal tubules, sodium reabsorption is increased directly by increased?**		
A-	sympathetic nerve stimulation of the kidney	B-	atrial natriuretic hormone secretion
C-	antidiuretic hormone secretion	D-	aldosterone secretion

13-	**Regarding water reabsorption?**
A-	The ascending limb of the loop of Henle is impermeable to water
B-	Maximum urine osmolarity is about 800 mosm/L
C-	The CD is freely permeable to water even in the absence of ADH
D-	The DCT is permeable to water

14-	**Reabsorption of Na?**		
A-	takes place in association with CL- & HCO3	B-	occurs only in PT
C-	is under control of parathormone hormone	D-	is a passive process

15-	**Which of the following substances will be more concentrated at the end of the proximal tubule than at the beginning of the proximal tubule?**						
A-	glucose	B-	sodium	C-	creatinine	D-	bicarbonate

16-	**The following substance leads to increased sodium reabsorption in kidney?**				
A-	Aldosterone	B-	Vasopressin 1	C-	ANP
D-	PgE2	E-	All of the above		

قَالَ الْفُضَيْلُ بْنُ عِيَاضٍ: بِقَدْرِ مَا يَصْغُرُ الذَّنْبُ عِنْدَكَ يَعْظُمُ عِنْدَ اللَّهِ، وَبِقَدْرِ مَا يَعْظُمُ عِنْدَكَ يَصْغُرُ عِنْدَ اللَّهِ.

17-	**Hydrogen ion secretion in proximal convoluted tubule uses?**		
A-	Passive diffusion	B-	Proton pump
C-	H+/K+ exchange	D-	H+/Na+ exchange

18-	**Glucose is reabsorped from apical membrane of proximal convoluted tubule using?**				
A-	Passive diffusion	B-	Endocytosis	C-	Facilitated diffusion
D-	Secondary active transport	E-	Primary active transport		

19-	**The characteristics of the Loop of Henle include?**
A-	Urine becomes more dilute as it reaches the tip of the loop
B-	Ascending limb is permeable to water
C-	In thin ascending limb urea is actively transported to interstitial fluid
D-	Na is reabsorbed by Na-K ATPase in as- cending limb

20-	**The functions of the renal proximal convoluted tubes include?**
A-	Reabsorption of all bicarbonate filtered at the glomerulus
B-	Production of renin
C-	Reabsorption of 80% of filtered bicarbonate
D-	Reabsorption of bicarbonate resulting in fixed acid secretion
E-	Hydrogen ion secretion by facilitated diffusion

21-	**For those substances that are actively reabsorbed or secreted, the highest limit of the amount that can be transported per unit time by the kidney tubules?**
A-	Depends on the rate at which the trans- port mechanism operates
B-	Is directly related to the plasma concentra- tion of the substance
C-	Depends on the tubular transport maximum
D-	Is dependent upon tubular load
E-	Depends on the GFR

22-	**Concerning urinary concentration?**
A-	the thick loop of Henle generates most of the osmotic
B-	gradient needed for reabsorption of water in the collecting duct
C-	in the absence of ADH, urine is not concentrated along the length of the collecting duct
D-	All of the above

23-	**The glucose started to appear in the urine before the transport maximum is?**
A-	Not all nephrons have the same transport maximum for glucose
B-	Filtered load of glucose is completely reabsorbed
C-	All nephrons have the same transport maximum
D-	None of the above

24-	**Most of the water reabsorption in the kidney is passes across the walls of the?**		
A-	Proximal tubules	B-	Ascending limb of the loop of Henle
C-	Descending limb of the loop of Henle	D-	Distal tubules

25-	Regarding tubular function?
A-	90% of the water is absorbed by the proximal convoluted tubule
B-	Renal threshold for glucose is 300mg/dl
C-	The main mechanism of the Na reabsorption f the tubular fluid to proximal convoluted tubules via Na/K/ATP pump
D-	The main mechanism of Na absorption in the ascending loop of Henle is via cotransport of Na/K/2C1

26-	**For those substances that are actively reabsorbed or secreted, the highest limit of the amount that can be transported per unit time by the kidney tubules?**
A-	Depends on the rate at which the transport mechanism operates
B-	Is directly related to the plasma concentration of the substance
C-	Depends on the tubular transport maximum
D-	Is dependent upon tubular load
E-	Depends on the GFR

27-	**Water reabsorption in the collecting duct?**
A-	Is dependent on aldosterone
B-	Is increased by ADH
C-	Has the largest percentage of total water reabsorbed
D-	Is facilitated by sodium reabsorption
E-	Independent of osmotic difference across the tubules

28-	**The transport of glucose across the luminal membrane in the proximal part of the renal tubule is?**
A-	Through insulin dependent glucose transporters
B-	Cannot take place against the concentration gradient
C-	Is not affected by Na+/K+pump inhibitors
D-	Is a co-transport with sodium
E-	Is decreased in diabetes mellitus

29-	**Which of the following is an action of parathyroid hormone (PTH) on the renal tubule?**
A-	Stimulation of adenylate cyclase
B-	Inhibition of distal tubule K+Secretion
C-	Inhibition of distal tubule Ca2+Reabsorption
D-	Stimulation of proximal tubule phosphate reabsorption
E-	Inhibition of production of 1,25-dihydroxycholecalciferol

30-	**As tubular fluid passes along a juxtamedullary nephron of a person with severe central diabetes insipidus and essentially no antidiuretic hormone, where is the osmolarity lowest?**
A-	Bowman's capsule (glomerular filtrate)
B-	Fluid leaving the proximal tubule and entering the loop of Henle
C-	Fluid leaving the descending thin limb and entering the Ascending thin limb of the loop of Henle
D-	Fluid leaving the thick ascending segment of the loop of Henle and entering the early distal tubule
E-	Fluid in the cortical collecting tubules
F-	Fluid leaving the collecting ducts (urine)

31-	**Which of the following occurs in type A intercalated cells of the collecting tubules?**
A-	Secretion of H+, reabsorption of HCO3 , and reabsorption of K+
B-	Secretion of H+, reabsorption of HCO3 , and secretion of K+
C-	Secretion of K+, reabsorption of Na+, and reabsorption of HCO3
D-	Reabsorption of H+, secretion of HCO3 , and secretion of K+
E-	Reabsorption of H+, secretion of HCO3 , and reabsorption of K+

32-	**Which of the following statements is incorrect?**
A-	Urea transporters UT-A1 and UTA-3 in the collecting ducts are activated by antidiuretic hormone (ADH)
B-	Urea reabsorption in the inner medullary collecting duct is greater than in the distal tubule during dehydration
C-	Increased ADH markedly increases urea reabsorption by the cortical collecting tubule
D-	The inner medullary collecting tubule reabsorbs more urea during antidiuresis than the thick ascending limb of Henle's loop
E-	The cortical collecting tubule is less permeable to urea than is The inner medullary collecting duct during antidiuresis
F-	Passive diffusion of urea into the thin loops of Henle is Facilitated by the urea transporter UT-A2

33-	**Atrial natriuretic peptide causes which of the following effects?**		
A-	Reduced renal tubular sodium reabsorption	B-	Reduced renin secretion
C-	Increased renal sodium excretion	D-	Only A and C
E-	A, B, and C		

عن أَنَسِ بن مالك – رضي الله عنه – قَالَ: قَالَ رَسُولُ اللَّهِ ﷺ:

«لَمَّا عُرِجَ بِي مَرَرْتُ بِقَوْمٍ لَهُمْ أَظْفَارٌ مِنْ نُحَاسٍ يَخْمِشُونَ وُجُوهَهُمْ وَصُدُورَهُمْ، فَقُلْتُ: مَنْ هَؤُلَاءِ يَا جِبْرِيلُ؟ فَقَالَ: هَؤُلَاءِ الَّذِينَ يَأْكُلُونَ لُحُومَ النَّاسِ وَيَقَعُونَ فِي أَعْرَاضِهِمْ» .

34-	**Because the usual rate of phosphate filtration exceeds the transport maximum for phosphate reabsorption, which statement is true?**
A-	All the phosphate that is filtered is reabsorbed
B-	More phosphate is reabsorbed than is filtered
C-	Phosphate in the tubules can contribute significantly to titratable acid in the urine
D-	The “threshold” for phosphate is usually not exceeded
E-	Parathyroid hormone must be secreted for phosphate reabsorption to occur

35-	**Which change would you expect to find in a dehydrated person deprived of water for 24 hours?**
A-	Decreased plasma renin activity
B-	Decreased plasma antidiuretic hormone concentration
C-	Increased plasma atrial natriuretic peptide concentration
D-	Increased water permeability of the collecting duct

36-	**Under conditions of normal renal function, which of the following statements is true of the concentration of urea in tubular fluid at the end of the proximal tubule?**
A-	It is higher than the concentration of urea in tubular fluid at the tip of the loop of Henle
B-	It is higher than the concentration of urea in the plasma
C-	It is higher than the concentration of urea in the final urine in antidiuresis
D-	It is lower than plasma urea concentration because of active urea reabsorption along the proximal tubule

37-	**Which statement is correct?**
A-	Urea reabsorption in the medullary collecting tubule is less than in the distal convoluted tubule during antidiuresis
B-	Urea concentration in the interstitial fluid of the renal cortex is greater than in the interstitial fluid of the renal medulla during antidiuresis
C-	The thick ascending limb of the loop of Henle reabsorbs more urea than the inner medullary collecting tubule during antidiuresis
D-	Urea reabsorption in the proximal tubule is greater than in the cortical collecting tubule

38-	**Which nephron segment is the primary site of magnesium reabsorption under normal conditions?**		
A-	Proximal tubule	B-	Descending limb of the loop of Henle
C-	Ascending limb of the loop of Henle	D-	Distal convoluted tubule
E-	Collecting ducts		

"سُئِلَ رَسولَ اللهِ ﷺ أَيُّ العَمَلِ أَحَبُّ إِلى اللهِ ؟ قالَ: أَدْوَمُهُ وإِنْ قَلَّ ."

39-	**Which of the following is true of the tubular fluid that passes through the lumen of the early distal tubule in the region of the macula densa?**
A-	It is usually isotonic
B-	It is usually hypotonic
C-	It is usually hypertonic
D-	It is hypertonic in antidiuresis
E-	It is hypertonic when the filtration rate of its own nephron decreases to 50% below normal

40-	**In a person on a high-potassium (200 mmol/day) diet, which part of the nephron would be expected to secrete the most potassium?**		
A-	Proximal tubule	B-	Descending loop of Henle
C-	Ascending loop of Henle	D-	Early distal tubule
E-	Collecting ducts		

41-	**Which change would tend to increase Ca2+ reabsorption in the renal tubule?**
A-	Extracellular fluid volume expansion
B-	Increased plasma parathyroid hormone concentration
C-	Increased blood pressure
D-	Decreased plasma phosphate concentration
E-	Metabolic acidosis

42-	**K+ excretion is markedly influenced by?**		
A-	Aldosterone	B-	Amount of Na+ delivered to tubules
C-	Rate of tubular secretion of H+	D-	All of the above

43-	**In the distal tubules, sodium reabsorption is increased directly by increased?**		
A-	Sympathetic nerve stimulation of the kidney	B-	Atrial natriuretic hormone secretion
C-	Antidiuretic hormone secretion	D-	Aldosterone secretion

44-	**Regarding water reabsorption?**
A-	The ascending limb of the loop of Henle is impermeable to water
B-	Maximum urine osmolarity is about 800 mosm/L
C-	The CD is freely permeable to water even in the absence of ADH
D-	The DCT is permeable to water

45-	**Hydrogen ion is NOT secreted in the following part of the nephron?**		
A-	Proximal convoluted tubule	B-	Loop of Henle

46-	**More hydrogen is secreted in?**		
A-	Alkalosis	B-	Administration of diamox
C-	Hypokalaemia	D-	Hyperventilation

47-	**Reabsorption of Na+?**		
A-	Takes place in association with CL- & HCO3-	B-	Occurs only in PT
C-	Is under control of parathormone hormone	D-	Is a passive process

48-	**The following substance leads to increased sodium reabsorption in kidney?**		
A-	Aldosterone	B-	Vasopressin
C-	ANP	D-	PgE2
E-	All of the above		

49-	**Which of the following substances will be more concentrated at the end of the proximal tubule than at the beginning of the proximal tubule?**		
A-	Glucose	B-	Creatinine
C-	Sodium	D-	Bicarbonate

50-	**Hydrogen ion secretion in proximal convoluted tubule uses?**		
A-	Passive diffusion	B-	Proton pump
C-	H+/K+ exchange	D-	H+/Na+ exchange

51-	**Glucose is reabsorped from apical membrane of proximal convoluted tubule using?**		
A-	Passive diffusion	B-	Endocytosis
C-	Facilitated diffusion	D-	Secondary active transport
E-	Primary active transport		

52-	**Most of the filtered water is reabsorped in?**		
A-	Proximal convoluted tubule	B-	Descending loop of Henle
C-	Ascending loop of Henle	D-	Distal convoluted tubule
E-	Collecting duct		

53-	**Which of the following statement is TRUE?**
A-	Thin ascending limb of loop of Heale has Na-K-2C1 active transport
B-	Thin descending limb of loop of Henle is impermeable to water
C-	Osmolality is greatest at the beginning of loop of Henle
D-	Water diffuse out of blood vessel from the descending limb of vasa recta into its ascending limb

54-	**Movement of urea in collecting duct is facilitated by?**		
A-	Noradrenaline	B-	Dopamine
C-	Vasopressin	D-	Angiotensin II
E-	All of the above		

55-	**The amount of potassium excreted by the kidney will decreases if?**		
A-	Distal tubular flow increases	B-	Circulating aldosterone level increase
C-	Dietary intake of potassium increase	D-	Na+ reabsorption by the distal nephron increases

56-	**The effect of antidiuretic hormone (ADH) on the kidney is to?**
A-	Increase the permeability of the distal nephron to water
B-	Increase the excretion of Na+
C-	Increase the excretion of water
D-	Increase the diameter of the renal artery

57-	**The ability of the kidney to excrete a concentrated urine will increase if?**
A-	The permeability of the proximal tubule to water decreases
B-	The rate of blood flow through the medulla decreases
C-	The rate of flow through the loop of Heale increases
D-	The activity of the Na-K pump in the loop of Henle decreases

58-	**Which of the following is most permeable to water?**		
A-	Thin ascending limb of the loop of Henle	B-	Distal convoluted tubule
C-	Thin descending limb of the loop of Henle	D-	Cortical portion of the collecting tubule

59-	**In the kidney?**
A-	Glucose is secreted by the collecting duct of PCT
B-	The renal threshold refers to glucose resorption from the urine PCT
C-	Glucose is reabsorbed by secondary active transport
D-	Glucose is reabsorbed in the loop of Henle

60-	**Renal acid secretion is affected by all of the following EXCEPT?**		
A-	PCO2	B-	K+
C-	Aldosterone	D-	Ca++

61-	**Renal threshold for glucose is?**		
A-	180 mg%	B-	300 mg%
C-	280 mg%	D-	120 mg %

62-	**At plasma concentrations of glucose higher than the renal threshold, the?**
A-	Clearance of glucose is zero
B-	Excretion rate of glucose equals the filtration rate of glucose
C-	Reabsorption rate of glucose equals the filtration rate of glucose
D-	Excretion rate of glucose increases with increasing plasma glucose concentration
E-	Reabsorption rate of glucose increases with increasing filtration rate

63-	**The renal transport maximum ™ for a substance is defined as the maximum?**
A-	Urinary excretion rate
B-	Tubular reabsorption or secretion rate
C-	Renal clearance rate
D-	Amount of a substance filtered by the glomeruli per minute

64-	**The renal threshold is the plasma level after which there is?**		
A-	Increase rate of solute excretion	B-	Beginning of solute excretion
C-	No solute excretion	D-	Filtration load less than tubular maximum

65-	**Sodium reabsorption?**
A-	Occurs in every segment of the nephron except the ascending limb of loop of Henle
B-	Be coupled with k+ and H+ excretion in the distal convoluted tubules
C-	In proximal convoluted tubules accounts for 90% of the filtered sodium

66-	**Most of the filtered Na+ is reabsorbed at?**		
A-	The proximal convoluted tubule	B-	The thick ascending limb of loop of Henle
C-	The distal convoluted tubule	D-	The collecting tubule

67-	**Sodium reabsorption?**
A-	Accounts for most of the energy requirement of the kidney
B-	At the luminal border of the cell is by active transport
C-	At the basolateral border of the cell is along both concentration and electric gradients
D-	In the descending limb of the loop of Henle is an active process

68-	**Sodium reabsorption in PCT?**		
A-	It needs a carrier but has no tubular maximum	B-	Occurs partially through the Para-cellular route
C-	If decreased it may lead to acidosis	D-	All of the above

69-	**Substances subjected to secondary active reabsorption in the proximal convoluted tubules includes?**		
A-	Urea	B-	Hydrogen ions
C-	Glucose	D-	All of the above

70-	**Bicarbonate?**
A-	Is subjected to tubular secretion in the proximal convoluted tubules
B-	Is responsible for concentration of urine
C-	Absorption depends on H secretion
D-	Absorption is linked to chloride reabsorption

71-	**Bicarbonate reabsorption in the proximal tubule?**		
A-	Is less than in distal tubule?	B-	Needs carbonic anhydrase enzyme
C-	Occurs by facilitated diffusion	D-	All of the above

72-	**In the descending limb of the loop of Henle?**
A-	Chloride ions are actively pumped
B-	Osmolarity of tubular fluids increases as it descends
C-	The wall is impermeable to water
D-	Glucose reabsorption is completed

73-	**K+ in the kidney is?**		
A-	Freely filtered, neither reabsorbed nor secreted	B-	Freely filtered, secreted but not reabsorbed
C-	Freely filtered, reabsorbed and secreted	D-	Freely filtered, reabsorbed and not secreted

74-	**In distal convoluted tubules?**
A-	Aldosterone increases K* reabsorption
B-	Aldosterone increases active water reabsorption
C-	Highly permeable to urea
D-	The wall is impermeable to water in absence of ADH

75-	**The tubular fluid becomes hypo-osmotic at the end of?**		
A-	Proximal convoluted tubule	B-	Descending limb of loop of Henle
C-	Ascending limb of loop of Henle	D-	Collecting tubules

76-	**In the thick ascending limb of the loop of Henle?**
A-	Cl- ions are reabsorbed passively through the paracellular route
B-	The osmolarity of the luminal fluid decreases as it ascends
C-	Water reabsorption is under the influence of ADH
D-	Glucose reabsorption is completed

77-	**Bicarbonate?**		
A-	Reabsorption stops in severe acidosis	B-	Reabsorption depends on Na reabsorption
C-	Secretion occurs in PCT	D-	Excretion in urine is inhibited in alkalosis

78-	**Most of the filtered water is reabsorbed at?**		
A-	Proximal convoluted tubule	B-	Descending limb of loop of Henle
C-	Medullary collecting ducts	D-	Collecting tubules

79-	**Water reabsorption?**
A-	Is linked to salt reabsorption in the distal tubule
B-	Is maximum at the descending limb of loop of Henle
C-	In the proximal tubule is under control of ADH
D-	None of the above

80-	**Factors affecting Na+ excretion includes?**		
A-	Diameter of afferent arteriole	B-	GFR
C-	Aldosterone hormone	D-	Atrial natriuretic peptide
E-	All of the above		

81-	**Sodium excretion is increased by?**		
A-	Aldosterone	B-	Calcitonin
C-	Epinephrine	D-	Estrogen
E-	Atrial natruritic peptide (ANP)		

82-	**Primary active transport of H+?**
A-	Occurs in PCT
B-	Occurs in Loop of Henle
C-	Is a function of all tubule cells
D-	Can secrete H+ against very high concentration gradient
E-	Accounts for 95% of total H + secretion

83-	**The "diluting segment" of the nephron is?**		
A-	PCT	B-	Descending limb of loop of Henle
C-	Ascending limb of loop of Henle	D-	Cortical collecting duct

84-	**The hyperosmolarity of renal medulla is due to increase content of?**		
A-	Na	B-	Glucose
C-	K	D-	Na' and urea

85-	**Concerning the function of the proximal tubule?**
A-	Most of the glomerular ultrafiltrate is reabsorbed in the proximal tubule in an iso-osmotic fashion
B-	The concentration of PAH doesn't change much along the length of the proximal tubule
C-	Under normal conditions, most of the filtered glucose and bicarbonate are reabsorbed in the proximal tubule
D-	A and C are correct

86-	**Within an hour following intravenous angiotensin II infusion, the following is(are) clinically evident?**
A-	Increased sodium uptake in the proximal tubular epithelium
B-	Systemic vasoconstriction
C-	Increased plasma aldosterone
D-	All are correct

87-	**Atrial natriuretic peptide?**		
A-	Increased H20 reabsorption	B-	Decreased sodium reabsorption
C-	All formation	D-	Increased sodium reabsorption

88-	**PTH?**		
A-	Increased H20 reabsorption	B-	Decreased sodium reabsorption
C-	All formation	D-	Decreased phosphate reabsorption

89-	**The followings are responsible for urine concentration EXCEPT?**		
A-	ADH	B-	Aldosterone
C-	Epinephrine	D-	Angiotensin II

90-	**Renin secretion is increased by the following EXCEPT?**		
A-	Hypovolemia	B-	Hyponatremia in DCT
C-	Sympathelic stimulation	D-	Hyponatremia in PCT

91-	**Acidification of urine takes place in?**		
A-	PCT	B-	DCT
C-	Collecting duct	D-	Ascending part of loop of Henle

92-	**Regarding Renal handling of water?**
A-	Water transport is restricted to PCT
B-	Water transport occurs by osmotic diffusion initiated by solute re- absorption
C-	Obligatory water re-absorption is under the control of ADH
D-	All of the above

93-	**For those substances that are actively reabsorbed, the maximal amount that can be transported per unit time by the kidney tubules?**
A-	Is termed the tubular transport maximum
B-	Requires specific transport systems for each substance transported
C-	Depends on the maximum rate at which the transport mechanism itself operates
D-	Is dependent upon tubular load

94-	What is the transport maximum for glucose (mg/min)?		
A-	100	B-	200
C-	300	D-	400

95-	At plasma glucose concentration of 400 mg/dl?
A-	Clearance of glucose is zero
B-	Excretion rate of glucose equal the filtration of glucose
C-	Reabsorption rate of glucose equals the filtration rate of glucose
D-	Excretion rate of glucose increases with further increasing of plasma concentration of glucose

96-	The renal glucose clearance in a healthy person is zero because glucose is?		
A-	Filtered and secreted	B-	Neither filtered nor secreted
C-	Filtered but not secreted	D-	Secreted but not filtered

97-	Regarding the regulation of plasma sodium?
A-	Granular cells of the afferent arterioles secrete renin when plasma sodium is low
B-	Angiotensin II is formed from renin by the action of an enzyme found on the endothelium of the pulmonary blood vessels
C-	The uptake of sodium ions is regulated by the proximal tubule
D-	Sodium transport by the thick ascending limb of the loop of Henle occurs by same mechanism as that of the proximal tubule

98-	The electrical difference between tubular lumen and proximal tubular cells?
A-	Equals zero mV
B-	Opposes glucose reabsorption
C-	Is sufficient to stop the active transport of Na+ under certain circumstances
D-	Favours influx of Na" through the luminal border into tubular epithelial cells

99-	Na' is reabsorbed from basolateral surface of the renal epithelial cells by?		
A-	Na+/H+ exchange	B-	Na-glucose cotransport
C-	Na-K pumNa-K pumpp	D-	Facilitated diffusion

100-	Na" ion reabsorption occurs mainly in?		
A-	The glomeruli	B-	The proximal convoluted tubule
C-	The distal convoluted tubule	D-	The loop of Henle

101-	Amino acids are almost completely reabsorbed from the glomerular filtrate via active transport in the?		
A-	Proximal convoluted tubule	B-	Loop of Henle
C-	Distal convoluted tubule	D-	Collecting duct

102-	**The collecting ducts in the kidney?**
A-	Can secrete water molecules activity into the urine
B-	Are responsible for most of the reabsorption of water that occurs in the kidneys
C-	Determine to a large extent the final osmolarity of urine
D-	Are rendered impermeable to water by anidiuretic hormone

103-	**Fluid entering the loop of Henle from the proximal convoluted tubule has higher?**		
A-	Osmolarity than plasma	B-	Na+ than plasma
C-	Glucose than plasma	D-	Urea than plasma

104-	**Within the distal convoluted tubule?**
A-	The osmolality of the tubular fluid is regulated by antidiuretic hormone
B-	Any filtered glucose is absorbed
C-	The osmolality of the tubular fluid is regulated by aldosterone
D-	The final composition of the urine is determined

105-	**Aldosterone?**		
A-	Mainly acts on the collecting ducts	B-	Promotes K+ excretion
C-	Favors metabolic acidosis	D-	Increases urinary osmolality

106-	**Antidiuretic hormone?**
A-	Decreases urinary osmolality through the osmotically driven absorption of H2O
B-	Is secreted from the adrenal cortex
C-	Is released via the posterior pituitary when plasma osmolality rises
D-	Is a steroid hormone
E-	Acts at intracellular receptors within the epithelium of the distal convoluted tubule

107-	**In the presence of ADH, the filtrate will be isotonic to plasma in the?**		
A-	Descending limb of the loop of Henle	B-	Ascending limb of the loop of Henle
C-	Cortical collecting tubule	D-	Medullary collecting tubule

108-	**The ability of the kidney to excrete concentrated urine will increase it?**
A-	The reabsorption of Na+ by the proximal tubule decreases
B-	The flow of filtrate through the loop of Henle increases
C-	The glomerular capillary pressure increases
D-	The permeability of the collecting duct to water increases

109-	**Regarding the control of water balance by the kidneys?**
A-	Cortex to the renal papilla
B-	ADH is secreted by the anterior pituitary in response to a decrease in the osmolality of the blood
C-	A normal person can't produce urine with osmolality greater than 300mOsmol/kg
D-	ADH acts on the P cells of collecting ducts to increase their permeability to water

110-	**Which of the following statements regarding secretion of hydrogen ions in renal tubules is correct?**
A-	The process of H+ secretion is coupled with regeneration of chloride
B-	Maximum H+ secretion occurs in the descending limb of the loop of Henle
C-	H secretion occurs by active transport
D-	H+ secretion results in secretion of equal moles of HC03- ions

111-	**The renal clearance of phosphate is increased by?**		
A-	Aldosterone	B-	Parathyroid hormone
C-	Norepinephrine	D-	Antidiuretic hormone (ADH)

112-	**A 40-year-old diabetic woman has glucosuria. This may occur due to inadequate glucose reabsorption at?**		
A-	Collecting duct	B-	Distal convoluted tubule
C-	Glomerulus	D-	Proximal convoluted tubule

113-	**Which one of the following comparisons between the distal nephron and the proximal tubule is correct?**
A-	The distal nephron is more permeable to hydrogen ion than the proximal tubule
B-	The distal nephron is less responsive to aldosterone than the proximal tubule
C-	The distal nephron secretes less potassium than does the proximal tubule
D-	The distal nephron has a more negative luminal potential than the proximal tubule

114-	**Distal convoluted tubules?**
A-	Forms part of the juxtaglomerular apparatus
B-	Is capable of reabsorbing H+ by a mechanism that involves carbonic anhydrase
C-	Is capable of secreting Na+ in exchange for K+
D-	Is capable of secreting Ca2+ in response to parathyroid hormone

115-	**The appearance of glucose in urine is a consequence of which of the following processes in the proximal tubule?**		
A-	Saturation of the Na+-glucose cotransporter	B-	Saturation of the Na+ - H+ exchanger
C-	Stimulation of glucose secretion	D-	Stimulation of glycogen breakdown

116-	Which of the following substances is actively secreted into the renal tubules?		
A-	Amino acids	B-	Chloride
C-	Sodium	D-	Potassium

117-	What percentage of the filtered load of sodium is reabsorbed by the proximal tubule?		
A-	15%	B-	25%
C-	45%	D-	65%

118-	Which of the following hormones produces decrease in urinary volume flow rate EXCEPT?		
A-	Angiotensin II	B-	Atrial natriuretic peptide
C-	Antidiuretic hormone	D-	Aldosterone

قال ابن رجب – رحمه الله – في جامع العلوم والأحكام:

(وأما الاستعانة بالله عز وجل دون غيره من الخلق، فلأن العبد عاجز عن الاستقلال بجلب مصالحه ودفع مضاره، ولا معين له على مصالح دينه ودنياه إلا الله عَمَل ، فمَن أعانه الله فهو المعان، ومَن خذله الله فهو المخذول، وهذا تحقيق قوله : (لا حول ولا قوة إلا بالله)، وهذه كلمة عظيمة، وهي كنز من كنوز الجنة، فالعبد محتاج إلى الاستعانة بالله في فعل المأمورات، وترك المحظورات، والصبر على المقدورات كلها؛ في الدنيا، وعند الموت، وبعده من أهوال البرزخ ويوم القيامة، ولا يقدر على الإعانة على ذلك إلا الله عَزَّوَجَلَّ، فمن حقق الاستعانة عليه في ذلك كله، أعانه، وفي الحديث الصحيح: احرص على ما ينفعك واستعن بالله ولا تعجز)

ANSWERS – Regulation of tubular reabsorption, renal handling of important substance									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	A	D	C	C	C	B	D	A	D
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
A	D	A	A	C	A	D	D	D	C
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
C	D	A	A	C	C	B	D	A	F
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
A	C	E	C	D	B	D	C	B	E
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
B	D	D	A	B	C	A	A	B	D
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
D	A	D	C	A	A	B	C	C	D
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**
A	D	B	B	B	A	A	D	D	C
71	**72**	**73**	**74**	**75**	**76**	**77**	**78**	**79**	**80**
B	B	C	D	C	B	B	A	B	E
81	**82**	**83**	**84**	**85**	**86**	**87**	**88**	**89**	**90**
E	D	C	D	D	D	B	D	C	B
91	**92**	**93**	**94**	**95**	**96**	**97**	**98**	**99**	**100**
A	B	A	C	D	C	B	D	C	B
101	**102**	**103**	**104**	**105**	**106**	**107**	**108**	**109**	**110**
A	C	D	A	B	C	C	D	D	C
111	**112**	**113**	**114**	**115**	**116**	**117**	**118**	**119**	**120**
B	D	C	A	A	D	D	B	-	-

❖ Acid base balance and Countercurrent system

1-	**All of the following are true of ammonium production EXCEPT?**
A-	It is mainly a function of PCT
B-	The source is mainly glutamine
C-	It represents the main mechanism in renal excretion of H+
D-	It is a low capacity high gradient system
E-	It is increased when the urine pH is low

2-	**The following is true about the Vasta Recta?**
A-	Descending vasta recta reabsorbs water from the interstitial fluid
B-	The descending capillaries loose Na and Urea
C-	Plasma becomes hyperosmotic as it descends in vasta recta
D-	Ascending capillaries loose water to interstitial fluid
E-	They are responsible for the countercurrent multiplier mechanism

3-	**With respect to the counter current system?**
A-	The loops of Henle act as counter current exchangers
B-	Solute diffuse out of vessels conducting blood towards the systemic veins
C-	Water diffuses out of the ascending vessels
D-	Counter current exchange is passive and can operate even if counter current multiplication ceases

4-	**Which of the following is a part of the counter current multiplier mechanism?**
A-	Active transport of NaCl in proximal convolute tubules
B-	Active transport of NaCl in collecting tubule
C-	Active transport of NaCl in ascending limb of the loop of Henle
D-	Active transport of NaCl in distal convoluted tubules

5-	**Counter current mechanism is a function of?**		
A-	Proximal convoluted tubule	B-	Loop of Henle
C-	Distal convoluted tubule	D-	All of the above

6-	**Countercurrent mechanism of the kidney consists of?**		
A-	PCT and collecting duct	B-	Loop of Henle and vasa recta
C-	Collecting tubule and vasa recta	D-	PCT and loop of Henle

"ما لا يُدرَكُ كُلُّه لا يُترَكُ جُلُّه".

7-	**A 45-year-old woman develops severe diarrhea while on vacation. She has the following arterial blood values:** **pH = 7.25** **PCO2 = 24 mm Hg** **[HCO3–] = 10 mEq/L** **Venous blood samples show decreased blood [K+] and a normal anion gap.** **The correct diagnosis for this patient is?**		
A-	Metabolic acidosis	B-	Metabolic alkalosis
C-	Respiratory acidosis	D-	Respiratory alkalosis
E-	Normal acid–base status		

8-	**From question above, Which of the following statements about this patient is correct?**
A-	She is hypoventilating
B-	The decreased arterial [HCO3–] is a result of buffering of excess H+ by HCO3-
C-	The decreased blood [K+] is a result of exchange of intracellular H+For extracellular K+
D-	The decreased blood [K+] is a result of increased circulating levels of aldosterone
E-	The decreased blood [K+] is a result of decreased circulating levels of Antidiuretic hormone (ADH)

9-	**The reabsorption of filtered HCO3- ?**
A-	Results in reabsorption of less than 50% of the filtered load when the plasma concentration of HCO3- is 24 mEq/L
B-	Acidifies tubular fluid to a pH of 4.4
C-	Is directly linked to excretion of H+ as NH4+
D-	Is inhibited by decreases in arterial PCO2
E-	Can proceed normally in the presence of a renal carbonic anhydrase inhibitor

10-	**A patient arrives at the emergency room with low arterial pressure, reduced tissue turgor, and the following arterial blood values:** **pH = 7.69** **[HCO3-] = 57 mEq/L** **PCO2 = 48 mm Hg** **Which of the following responses would also be expected to occur in this patient?**		
A-	Hyperventilation	B-	Decreased K+secretion by the distal tubules
C-	Increased ratio of H2PO4 – to HPO4–2 in urine	D-	Exchange of intracellular H+for extracellular K+

فَاعْكُفْ عَلَى الْعِلْمِ تَبْلُغْ شَأْوَ مَنْزِلَةٍ *** فِي الْفَضْلِ مَحْفُوفَةٍ بِالْعِزِّ وَالْكَرَمِ

فَلَيْسَ يَجْنِي ثِمَارَ الْفَوْزِ يَانِعَةً *** مِنْ جَنَّةِ الْعِلْمِ إِلَّا صَادِقُ الْهِمَمِ

11-	**Which of the following is a cause of metabolic alkalosis?**		
A-	Diarrhea	B-	Chronic renal failure
C-	Ethylene glycol ingestion	D-	Treatment with acetazolamide
E-	Hyperaldosteronism		
F-	Salicylate poisoning		

12-	**Which set of arterial blood values describes a patient with partially compensated respiratory alkalosis after 1 month on a mechanical ventilator?**
A-	PH, 7.65, HCO3 (mEq/L),48, Pco2(mm Hg), 45
B-	PH, 7.50,HCO3 (mEq/L), 15, Pco2(mm Hg), 20
C-	PH, 7.40,HCO3 (mEq/L),24, Pco2(mm Hg), 40
D-	PH, 7.32,HCO3 (mEq/L),30, Pco2(mm Hg), 60
E-	PH, 7.31,HCO3 (mEq/L), 16,Pco2(mm Hg), 33

13-	**Which set of arterial blood values describes a patient with chronic renal failure (eating a normal protein diet) and decreased urinary excretion of NH4+?**
A-	PH, 7.65, HCO3 (mEq/L),48, Pco2(mm Hg), 45
B-	PH, 7.50, HCO3 (mEq/L), 15, Pco2(mm Hg), 20
C-	PH, 7.40, HCO3 (mEq/L),24, Pco2(mm Hg), 40
D-	PH, 7.32, HCO3 (mEq/L),30, Pco2(mm Hg), 60
E-	PH, 7.31, HCO3 (mEq/L), 16, Pco2(mm Hg), 33

14-	**A female graduate student is hyperventilating prior to her oral comprehensive examination. She is light-headed, and her feet and hands are numb and tingling. Which of the following set of blood values would be observed in the emergency room?**
A-	PH, 7.3, Pco2 (mm Hg),30, Po2(mm Hg), 100,ionized Ca2+, Decreased
B-	PH, 7.3, Pco2 (mm Hg),50, Po2(mm Hg), 90,ionized Ca2+, increased
C-	PH, 7.4, Pco2 (mm Hg),40, Po2(mm Hg), 100,ionized Ca2+, Normal
D-	PH, 7.5, Pco2 (mm Hg),30, Po2(mm Hg), 110,ionized Ca2+, Decreased
E-	PH, 7.5, Pco2 (mm Hg),50, Po2(mm Hg), 90,ionized Ca2+, Increased

15-	**Acute metabolic alkalosis tends to _____K+ secretion by the cortical collecting tubules and _____ plasma K+concentration.**		
A-	Decrease, decrease	B-	Decrease, increase
C-	Increase, increase	D-	Increase, decrease
E-	Cause no change in, increase		
F-	Cause no change in, cause no change in		

16-	**Which of the following would tend to cause hypokalemia by shifting potassium from the extracellular fluid into the intracellular fluid?**		
A-	Metabolic alkalosis	B-	Insulin deficiency
C-	Aldosterone deficiency	D-	Beta-adrenergic receptor blockade
E-	Increased extracellular fluid osmolarity		

17-	**When the dietary intake of K+increases, body K+ balance is maintained by an increase in K+ excretion primarily by which of the following?**
A-	Decreased glomerular filtration of K+
B-	Decreased reabsorption of K+ by the proximal tubule
C-	Decreased reabsorption of K+by the thick ascending limb of the loop of Henle
D-	Increased K+ secretion by the late distal and collecting tubules
E-	Shift of K+into the intracellular compartment

18-	**A 65-year-old man had a heart attack and experiences cardiopulmonary arrest while being transported to the emergency department. Use the following laboratory values obtained from arterial blood to answer the question.** **Plasma pH = 7.12** **Plasma Pco2 = 60 mm Hg** **Plasma HCO3 concentration = 19 mEq/l** **Which of the following options best describes his acid-base disorder?**
A-	Respiratory acidosis with partial renal compensation
B-	Metabolic acidosis with partial respiratory compensation
C-	Mixed acidosis: combined metabolic and respiratory acidosis
D-	Mixed alkalosis: combined respiratory and metabolic alkalosis

19-	**From question above, In this patient, which of the following laboratory results would be expected, compared with normal?**		
A-	Increased renal excretion of bicarbonate(HCO_3^-)	B-	Decreased urinary titratable acid
C-	Increased urine pH	D-	Increased renal excretion of ammonia (NH_4^+)

20-	**Which change would you expect to find in a patient who developed acute renal failure after ingesting poisonous mushrooms that caused renal tubular necrosis?**
A-	Increased plasma bicarbonate concentration
B-	Metabolic acidosis
C-	Decreased plasma potassium concentration
D-	Decreased blood urea nitrogen concentration
E-	Decreased hydrostatic pressure in Bowman's capsule

21-	**In a person with normal kidneys and normal lungs who has chronic metabolic acidosis, you would expect to find all of the following, compared with normal, EXCEPT?**		
A-	Increased renal excretion of NH4Cl	B-	Decreased urine pH
C-	Decreased urine HCO3 excretion	D-	Increased Plasma HCO3 concentration
E-	Decreased plasma Pco2		

22-	**In a person with chronic respiratory acidosis who has partial renal compensation, you would expect to find which changes, compared with normal? _____ urinary excretion of NH4+; _____ plasma concentration; and _____ urine pH.**		
A-	Increased, increased, decreased	B-	Increased, decreased, decreased
C-	No change in, increased, decreased	D-	No change in, no change in, decreased
E-	Increased, no change in, increased		

23-	**Movement of urea in collecting duct is facilitated by?**		
A-	Noradrenaline	B-	Dopamine
C-	Vasopressin	D-	Angiotensin II
E-	All of the above		

24-	**The effect of antidiuretic hormone (ADH) on the kidney is to?**
A-	Increase the permeability of the distal nephron to water
B-	Increase the excretion of Na+
C-	Increase the excretion of water
D-	Increase the diameter of the renal artery

25-	**In the renal tubules, bicarbonate is responsible for buffering hydrogen ion in?**		
A-	Proximal convoluted tubule	B-	Loop of Henle
C-	Distal convoluted tubule	D-	Collecting ducts

26-	**Which of the following best describes the changes in uncompensated respiratory alkalosis?**		
A-	Low pH, HCO3 and PaCO2	B-	High pH. Low HCO and PaCO2
C-	Low pH and HCO3 and normal PaCO2	D-	High pH, low HCO3 and normal PaCO2

27-	**In the normal bladder micturition is?**		
A-	Intiated by the pelvic nerves	B-	Initiated at a volume of 600 mL
C-	Significantly affected by sympathetic nerves	D-	Not facilitated at the level of the brain stem

"ولا ينالُ المعالي الغُرّ غير فتىً *** يدوسُ شوكَ العوالي غيرَ مُنتعلِ"

28-	**In the distal convoluted tubule?**
A-	Water reabsorption is linked to solute reabsorption
B-	Intercalated cells secrete H and K
C-	Intercalated B cells secrete H during acidosis
D-	The main mechanism for hydrogen ion secretion is by primary active transport

29-	**Tubular secretion of hydrogen ions is?**		
A-	Increased in renal failure	B-	Increased by lowering P Co2 in the blood
C-	Increased in cases of respiratory failure	D-	Increased by hyperventilation

30-	**Metabolic acidosis?**
A-	Accompanied by increase in plasma bicarbonate
B-	Compensated for by decreased rate of respiration
C-	Occurs in cases of severe vomiting
D-	The main cause is hyperventilation
E-	Compensated for by hyperventilation

31-	**In metabolic acidosis all of the following are correct EXCEPT?**
A-	Increase H concentration in blood and tissue fluids
B-	Decreased CO2 in blood
C-	Hypoventilation
D-	Decreased blood bicarbonate

32-	**Respiratory acidosis?**
A-	Occurs due to hyperventilation
B-	Occurs during muscular exercise due to accumulation of lactic acid
C-	Bicarbonate in blood may be high or normal
D-	Is compensated for by excretion of bicarbonate in urine

33-	**Bicarbonate buffer system is very effective in neutralizing the acid-base disturbances in the body because?**		
A-	Qantitavely too large	B-	Its pKa is very near to the pH
C-	Easily regulated and replaced	D-	All of the above

34-	**Destruction of the sensory nerve fibers from the urinary bladder to the spinal cord results in?**		
A-	Failure of micturition reflex	B-	Complete loss of control on bladder
C-	Bladder empties by overflow dribbling	D-	All of the above

35-	Renal ammonium ion excretion is increased by aldosterone, elevated blood pCO2 and acidemia?		
A-	True	B-	False

36-	Respiratory alkalosis occurs in?		
A-	Diabetic ketoacidosis	B-	Pyloric stenosis
C-	Primary hyper aldosteronism	D-	Hyperventilation

37-	Respiratory acidosis is caused by all of the following EXCEPT?		
A-	Diabetic keto-acidosis	B-	Breath holding
C-	Respiratory distress syndrome	D-	Pneumonia

38-	Metabolic alkalosis will be observed in a patient with?		
A-	Hyperaldosteronism	B-	Hyperventilation
C-	Persistent diarrhea	D-	Renal failure

39-	With respect to the renal regulation of acid-base balance, one of the following is incorrect?
A-	Ammonium ions are mainly produced in the loop of Henle
B-	Glutamine metabolism by the kidneys results in bicarbonate production
C-	Ammonia production by the kidneys is increased in acidosis
D-	Secreted hydrogen ions are buffered by the phosphate buffer system in tubular fluid

40-	Metabolic alkalosis occurs due to?		
A-	Vomiting	B-	Starvation
C-	Heart failure	D-	Hyperparathyroidism

41-	Which one of the following statements about ammonia (NH3) is correct?
A-	It is impermeable through the epithelial cells of the proximal tubule
B-	It is classified as a titratable acid
C-	It is produced by epithelial cells in the distal nephron
D-	Its synthesis is increased in respiratory acidosis

42-	The most important buffer in the ECF and plasma is:		
A-	Bicarbonate buffer.	B-	Phosphate buffer.
C-	Protein buffer.	D-	Hemoglobin

43-	Micturition reflex is?
A-	Absent in children
B-	its Center in the medulla
C-	Can be easily voluntarily inhibited if the urinary bladder volume reaches 150 ml
D-	Efferent fibers are sympathetic only

44-	**In the normal bladder, micturition is?**		
A-	Initiated by the pelvic nerves	B-	Significantly affected by sympathetic nerves
C-	Initiated at a volume of 600m.L	D-	Not facilitated at the level of the brain

ANSWERS – Acid base balance and Countercurrent system									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	C	B	C	B	B	A	E	D	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
F	B	E	D	D	A	D	C	D	B
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
D	A	C	A	A	B	A	C	C	E
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
C	C	C	D	A	D	A	A	A	A
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
D	A	C	A	-	-	-	-	-	-

"قُلْ لِلَّذي هجرَ الكِتابَ وما تَلا
في يَومِهِ وِردًا مِنَ القُرآنِ
أتُراكَ تُبصِرُ لِلسَّعادَةِ مَوطِنًا ؟
أمْ تاهَ قلبُكَ في دُجى الأحزانِ !"

Microbiology

❖ MICROBIOLOGY

1-	**A sexually active woman develops dysuria, pyuria, and fever suggestive of urinary tract infection. Urine cultures show G+ bacteria in clusters that are catalase positive and coagulase negative. Which of the following would of the following most likely cause of the cystitis?**		
A-	Viridans Streptococci	B-	Streptococcus pyogenes
C-	Staphylococcus aureus	D-	Staphylococcus epidermidis
E-	Staphylococcus saprophyticus		

2-	**Urinary tract infection is not caused by?**		
A-	Escherichia coli	B-	Klebsiella pneumonia
C-	Pseudomonas aeruginosa	D-	Proteus mirabilis
E-	Shigella dysenterica		

3-	**What are the symptoms of cystitis?**		
A-	Frequency	B-	Hematuria
C-	Dysuria	D-	Urgency
E-	All of above		

4-	**All of the following statements are true regarding E. Coli, EXCEPT?**
A-	Coli is the most common cause of urinary tract infection
B-	Urinary tract infection can result in bacteremia with clinical signs of sepsis
C-	Most of the urinary tract infections are caused by uropathogenic E coli
D-	These organisms produce hemolysin, which facilitates tissue invasion
E-	In urinary tract infections with E. coli. the urine becomes alkaline. Promoting stone formation

5-	**All statements are true regarding Proteus species, EXECPT?**
A-	Proteus Vulgaris is an important nosocomial pathogen
B-	Proteus species produce urease
C-	In urinary tract infections with Proteus species, stone formation is promoted
D-	Proteus species are non-motile
E-	proteus species produce bacteremia, pneumonia, and focal lesions in debilitated patients

6-	**All of the following statements are true regarding Klebsiella, EXCEPT?**
A-	Klebsiella are gram-negative bacilli which are thick and short
B-	They are non-motile and tend to produce large capsules on primary isolation
C-	This organism grows well on ordinary nutrient media
D-	Colonies are large. raised, moist and non-mucoid
E-	They give rise to pink colored colonies

7-	**Which of the following is oxidase positive?**		
A-	E coli	B-	K pneumoniae
C-	P. vulgaris	D-	Salmonella
E-	P. aeruginosa		

8-	**All of the following organisms can be found in the urine except?**		
A-	Coli	B-	Candida
C-	H. influenza	D-	Enterococci
E-	S. Saprophyticus		

9-	**Which of the following statements is false about asymptomatic bacteriuria?**
A-	Asymptomatic bacteriuria is very common in clinical practice
B-	Most patients with asymptomatic bacteriuria will never develop symptomatic urinary tract infections
C-	Most patients will have no adverse consequences from asymptomatic bacteriuria
D-	Most patients with asymptomatic bacteriuria benefit from treatment
E-	Obstructive uropathy is thought to increase the likelihood of asymptomatic bacteriuria

10-	**All of the followings are risk factors of UTI, Except?**
A-	Males are more likely to get urinary tract infections, due to the closeness of urethra to the anal opening
B-	Sexually active women
C-	Honeymoon cystitis during early marriage
D-	Senile prostatic hypertrophy
E-	People who have diabetes mellitus and pregnant women

11-	**All of the following are correct regarding CAUTI, Except?**
A-	CAUTI are the most common infections in health care facilities
B-	The placement of a urinary catheter into the bladder disrupts the natural host defense mechanisms
C-	The urinary catheter helps the complete elimination of bladder urine without a residual pool of stagnant urine
D-	causative bacteria of CAUTI can be from the patient's own skin and urogenital flora
E-	The biofilm also acts to protect the bacteria from the host immune system

12-	**All of the following statements are true regarding diagnosis of UTI, Except?**
A-	It is generally accepted that 10^3 CEU/mL of urine is significant bacteriuria
B-	Some young women with acute dysuria suggesting cystitis have negative urine cultures for bacteria
C-	Urine analysis, looking for the presence of urinary nitrites, pus cells
D-	Urine analysis, looking for the presence of leukocyte esterase, presence of red blood cells, or bacteria
E-	Culture and sensitivity of urine

13-	**Enterococcus faecalis is?**
A-	A frequent cause of healthcare-associated infections
B-	Usually treated by a single antibiotic in cases of bacteremia
C-	Unable to grow in the presence of bile and NaCl
D-	A member of Lancefield group B

14-	**Staphylococcus epidermidis can be distinguished from Staphylococcus saprophyticus by?**		
A-	Gram staining	B-	Novobiocin sensitivity
C-	Coagulase production	D-	Production of haemolysis on blood agar

15-	**Proteus spp. Are?**		
A-	Motile pleomorphic gram negative bacilli	B-	Lactose fermenter (LF)
C-	Urease-negative	D-	Phenylalanine deaminase-negative

16-	**All of the followings are characters of Pseudomonas aeruginosa?**
A-	Gram negative bacilli
B-	Oxidase negative
C-	Resistance to high concentration of salt and dye
D-	Don't ferment carbohydrate

17-	**both Acinetobacter and p.aeruginosa have high antibiotic resistance?**		
A-	True	B-	False

18-	**All able to withstand disinfection of the following are characteristics of Pseudomonas aeurginosa EXCEPT?**		
A-	Able to withstand disinfection	B-	Able to grow at high temperature
C-	Oxidase positive	D-	Ferment glucose

19-	**Sterile pyuria?**
A-	Caused by renal tuberculosis
B-	Is defined as presence of pus in urine in absence of bacterial growth
C-	Intake of antibiotics before urine sampling is considered as one of the major causes
D-	All of the above

20-	**All of the following are characters of enterococci Except?**
A-	Grow in bile esculin media
B-	Transmitted mainly in hospitals by inhalation
C-	Can grow at wide range of temperature 10-45°C
D-	Cause meningitis and bacteremia in neonates

21-	**Alkaline urine promotes which type stone formation?**		
A-	Cystine stones	B-	Calcium phosphate stones
C-	Struvite stones	D-	Both B & C

22-	**which of the following is not from predisposing factors of UTI?**		
A-	BPH	B-	Structural abnormalities of UT
C-	Shortness of urethra	D-	None of the above

23-	**The main cause of nosocomial UTI is diabetes?**		
A-	True	B-	False

24-	**All of the following cause UTI except.....?**		
A-	Proteus	B-	Pseudomonas
C-	Adenovirus	D-	None of the above

25-	**In viable count process , we must centrifuge the sample before spread it on media?**		
A-	True	B-	False

26-	**A sample collected from a patient and applied for bacterial count the result is 10,000 of staph.aureus this result indicate.....?**		
A-	UTI	B-	UT contamination
C-	None of the above		

27-	**Patient diagnosed as sterile pyuria, which of the following may explain his condition?**		
A-	He may has renal TB	B-	He use antibiotic for his condition
C-	He may be infected by ureaplasma	D-	All of the above

28-	**The difference diabetic patient and other one suffer from cystitis is that the amount of urine is.....in diabetic patient?**		
A-	More	B-	Less
C-	Equal but without pain	D-	There is no difference

29-	**IMVIC test of E coli is?**		
A-	++--	B-	--++
C-	+-+-	D-	-+-+

30-	**Staph.saprophyticus is novobioicin sensitive?**		
A-	True	B-	False

31-	**After spreading a sample of enterococci on bile-esculine media it gives +ve result ,this patient can be treated by.....?**		
A-	Cephadroxil	B-	Penicillin
C-	Vancomycin	D-	None of the above

32-	**The most common cause of UTI in the community is...?**		
A-	Enterobacter	B-	S. aureus
C-	E.coli	D-	Klebsiella

33-	**A young woman presents with recurrent urinary tract infections caused by the same Proteus mirabilis strain. What is the major concern?**
A-	She does not take her medication
B-	She is pregnant as pregnant patients are more susceptible to UTIs
C-	She has a bladder or kidney stone
D-	Her partner is infected

34-	**Complications of UTI include the followings EXCEPT?**		
A-	Recurrent infections	B-	Urethral stricture
C-	Permanent kidney damage	D-	Vesico-ureteric reflux

35-	**It is one of the virulence factors of E.coli?**		
A-	Fimbriae	B-	IgA protease
C-	Elastase	D-	Phospholipase

36-	**Biochemical reactions of E.coli are the following except?**		
A-	Positive Indole test	B-	Negative Citrate
C-	Positive Methyle Red test (MR)	D-	Positive oxidase

37-	**The most common bacterial cause of UTI is?**		
A-	E.coli	B-	Klebsiella
C-	Neisseria gonorrhea	D-	Mycobacterium tuberculosis

38-	**IMVC of Klebsiella pnemoniae is?**		
A-	++--	B-	--++
C-	---+	D-	+-+-

39-	**The causes of sterile pyuria are ?**		
A-	Mycobacterium tuberculosis	B-	Anaerobes
C-	Antibiotic therapy	D-	All of the above

40-	All the following are true about proteus sp. Except?		
A-	Causative agent of UTI	B-	Gram positive bacilli
C-	Urease positive	D-	Characteristic swarming growth

41-	Staphylococcus saprophyticus is ?
A-	Gram negative bacilli
B-	Catalase -ve
C-	Sensitive to novobiocin
D-	Common cause of UTI among sexually active young females

42-	All the following regarding UTI are true except?
A-	UTIs are common, especially among women
B-	Streptococcus pyogenes is the commonest cause and accounts >80% of infections
C-	UTIs in men are less common and primarily occur after 50 years of age
D-	UTIs occur in two general settings: community acquired and hospital (nosocomialy) acquired

43-	Diseases caused by E-coli are?		
A-	Community and hospital acquired UTI	B-	Neonatal meningitis
C-	Diarrhea	D-	All of the above

44-	Urethritis is caused by all except?		
A-	Gonococci	B-	Staphylococcus aureus
C-	Mycoplasma genitalium	D-	Ureaplasma urealyticum

45-	Risk factors for acute pyelonephritis?		
A-	Disruption of urine flow	B-	Vesicoureteral reflux
C-	Indwelling urinary catheter	D-	All of the above

46-	Significant bacteriuria is?		
A-	>= 10.000 bacteria/ml	B-	>= 100.000 bacteria/ml
C-	>= 1000 bacteria/ml	D-	None of the above

47-	Enterobacter is similar to klebsiella except in being?		
A-	Indole negative	B-	Methyl red negative
C-	Motile	D-	Citrate positive

48-	Hospital acquired UTI is most commonly caused by except?		
A-	P. aeruginosa	B-	Staph. Saprophyticus
C-	Klebsiella	D-	Entero. faecalis

49-	Acute hemorrhagic cystitis in children is caused by?		
A-	CMV	B-	HSV-2
C-	HPV	D-	Adenoviruses

50-	Which is the most common route of bacterial entry to the urinary tract?		
A-	Ascending infection	B-	Blood borne spread
C-	Lymphatic spread	D-	Direct extension from other organs

51-	A 60 year-old woman with uti the organisms isolated are gram-positive cocci that are catalase-negative and capable of growth in 6.5% sodium chloride. Which of the following is the most likely causal agent?		
A-	Staphylococcus aureus	B-	Staphylococcus saprophyticus
C-	Enterococcus faecalis	D-	Streptococcus pyogenes

52-	A 45 year-old woman presents to the emergency department with intense pain in her lower back and a burning sensation upon urination. A urine culture was taken and plated on MacConkey agar. Gram-negative rods that did not ferment lactose were identified. Which virulence factor of the causal agent is most important to pathogenesis?						
A-	Coagulase	B-	Urease	C-	Catalase	D-	Exotoxin

53-	Precautions during collection of urine sample for laboratory diagnosis in case of UTI include?		
A-	Cleaning and drying of external genitalia	B-	Middle urine sample is needed
C-	Collection in sterile container	D-	All of the above

54-	Sterile pyuria means?
A-	Presence of pus in urine (more than 10 cells / HPF)
B-	Presence of pus in urine in absence of bacterial growth on ordinary media
C-	Presence of pus in urine with bacterial growth on ordinary media

55-	Sterile pyuria can occur with?		
A-	Renal tuberculosis	B-	Anaerobes
C-	Mycoplasma	D-	All of the above

56-	Urine cultures are specially indicated in the following situations?		
A-	Complicated UTI including pyelonephritis	B-	Transplant and diabetic patients
C-	Pregnancy	D-	All of the above

57-	Bacterial count 100.000 bacteria per ml of urine or more?				
A-	Indicates UTL	B-	Regarded as contamination	C-	Is of no significance

58-	**Bacterial count less than 10.000 bacteria per ml of urine?**				
A-	Indicates UTI	B-	Regarded as contamination	C-	Is of no significance

59-	**A count of 10.000 organisms per ml of urine is considered significant if?**
A-	The organism is gram positive and only one type
B-	The organism is gram positive and many types
C-	The organism is gram negative and many types

60-	**Antibiotic sensitivity test is especially indicated in case of?**
A-	Urine with significant bacteriuria , particularly from patient with recurrent UTI
B-	Complicated UTI
C-	A and B
D-	None of the above

61-	**On Macconkey medium E.coli gives?**				
A-	Pale colonies	B-	Metallic green colonies	C-	Rose pink colonies
D-	None of the above				

62-	**On EMB E.coli gives?**				
A-	Pale colonies	B-	Metallic green colonies	C-	Pink colonies
D-	None of the above				

63-	**On triple sugar iron (TSI) E.coli gives**		
A-	Red slant and red butt	B-	Yellow slant and yellow butt
C-	Yellow slant and red burt	D-	None of the above

64-	**On Macconkey medium proteus gives?**		
A-	Metallic green colonies	B-	Rose pink colonies
C-	Pale colonies	D-	None of the above

65-	**Proteus gives positive?**				
A-	Urease test	B-	H2S production test	C-	A and B
D-	None of the above				

66-	**On triple sugar iron (TSI) proteus gives?**		
A-	Yellow slant and yellow butt	B-	Yellow slant and red butt
C-	Red slant and black butt	D-	None of the above

﴿لِيَجْزِيَهُمُ اللَّهُ أَحْسَنَ مَا عَمِلُوا وَيَزِيدَهُم مِّن فَضْلِهِ﴾

67-	**The following bacteria can grow on bile esculin azide?**		
A-	Enterococci	B-	Staphylococcus aureus
C-	Chlamydia	D-	None of the above

68-	**On bile esculin azide enterococci gives?**		
A-	Black colonies	B-	Rose pink colonies
C-	Pale colonies	D-	None of the above

69-	**A 21 year newly married woman developed a urinary tract infection (UTI) . At the time she sought medical advice , she was febrile and complained of painful urination and flank pain . Her urine appeared " cloudy " . Urine culture yields a lactose - fermenting , indole - positive , Gram negative bacillus . The infectiveness of the organism responsible for this urinary tract infection is associated with specific?**				
A-	exotoxins	B-	K antigens	C-	Metabolic properties
D-	P fimbriae	E-	Plasmids		

70-	**Which is a common UTI risk factor in adults?**		
A-	Enlarged prostate	B-	Catheter usage
C-	Diabetes	D-	All of the above

71-	**What are signs and symptoms of a UTI in adults?**		
A-	Frequent urge to urinate	B-	Pain during urination
C-	Milky / cloudy urine	D-	All of the above

72-	**Bacteria from the intestines (most commonly) often infect the urinary tract by ascending from the perineal area into the lower urinary tract?**		
A-	Escherichia coli	B-	Bacillus anthracis
C-	Staphyloccus aureus	D-	Streptococcus pyogenes , Group A

73-	**Infections are among the most common urinary tract infections . (multiple response)?**		
A-	Urethritis	B-	Prostatitis
C-	Cystitis	D-	Nosocomial (hospital acquired)

74-	**A 40-year-old women has dysuria, urine analysis showed an Alkaline urine and infection with a urea splitting organism, What is the type of stone?**						
A-	Phosphate	B-	Struvite	C-	Calcium stone	D-	Uric acid

75-	**Sterile pyuria can be caused by all except?**		
A-	Streptococcus saprophyticus	B-	Mycoplasma hominis
C-	Chlamydia	D-	Mycobacterium Tb

ANSWERS – Microbiology									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
E	E	E	D	D	D	E	C	B	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	A	A	B	A	B	A	D	D	B
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
D	D	B	D	B	A	D	A	A	B
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
D	C	C	D	A	D	A	B	D	B
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
D	B	D	B	D	A	C	B	D	A
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
C	B	D	B	D	D	A	B	A	C
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**
C	B	B	C	C	C	A	A	C	D
71	**72**	**73**	**74**	**75**	**76**	**77**	**78**	**79**	**80**
D	A	C&D	B	A	-	-	-	-	-

(اختيارُ الراحةِ الآنَ لن يورثَ إلا الشقاءَ لاحقًا).

يقولُ ابنُ القيّمِ –رحمهُ اللّهُ:-

"قد أجمعَ عقلاءُ كلِّ أمةٍ على أنّ النعيمَ لا يُدرَكُ بالنعيمِ، وأنّ من آثرَ الراحةَ فاتتهُ الراحةُ، وأنه بقدرِ الأهوالِ واحتمالِ المشاقِّ تكونُ الفرحةُ واللذةُ؛ فلا فرحةَ لمن لا همَّ له، ولا لذةَ لمن لا صبرَ له، ولا نعيمَ لمن لا شقاءَ له، ولا راحةَ لمن لا تعبَ له".

Parasitology

❖ Parasitology

1-	**Man acts as a blind intermediate host in which one of the following cestodes?**		
A-	Echinococcus granulosus	B-	Diphyllobothrium latum
C-	Hymenolepis nana	D-	Taenia saginata

2-	**The Schistosoma haematobium deposit their eggs in the?**		
A-	Small venules in Small intestinal wall	B-	Small venules in large intestinal wall
C-	Small venules in bladder wall	D-	Small venules of biliary tracts

3-	**Genitourinary flagellates are represented by?**		
A-	Giardia lamblia	B-	Toxoplasma gondii
C-	Plasmodium species	D-	Trichomonas vaginalis

4-	**Which one of the following Parasites can cause splenomegaly?**		
A-	Schistosoma	B-	Hymenolepis nana
C-	Onchocerca volvulus	D-	Ascaris lumbercoides

5-	**Which one of the following immunopathological reactions produced due to infection with Schistomiasis?**		
A-	Immune complex deposition	B-	Granuloma formation
C-	Immune suppression	D-	Allergic reactions

6-	**Granuloma formation is a form of?**		
A-	Type 1 hypersensitivity reaction	B-	Type 2 hypersensitivity reaction
C-	Type 3 hypersensitivity reaction	D-	Type 4 hypersensitivity reaction

7-	**Which one of the following Parasites can cause microcytic hypochromic anaemia?**		
A-	Diphyllobothrium latum	B-	Leishmania donovani
C-	Plasmodium spp	D-	Schistosoma spp

8-	**Babesia can affect renal system by causing?**		
A-	Renal failure	B-	Hydronephrosis
C-	Carcinoma	D-	Stones
E-	None of the above		

9-	**Leishmania donovani can affect renal system by?**		
A-	Causing renal amyloidosis	B-	Proteinuria
C-	Membranoproliferative glomerulonephritis	D-	All of the above
E-	None of the above		

10-	**All are true except?**
A-	Only two of the malaria parasites, namely, P. malariae and P. falciparum. Are associated with renal disease
B-	Plasmodium malariae can cause nephrotic syndrome and acute renal failure
C-	Plasmodium falciparum can cause acute renal failure
D-	Plasmodium falciparum cause renal failure due to impaired blood flow in the microcirculation due to adhesiveness of erythrocytes and hypovolemia
E-	None of the above

11-	**All the following parasites can affect renal system except?**		
A-	Schistosoma haematobium	B-	Schistosoma mansoni
C-	Leishmania major	D-	African trypanosome
E-	Leishmania donovani		

12-	**All the following statements are true except?**
A-	Filarisis can affect renal system by immune complex formation
B-	Filarisis can affect renal system causing nephrotic syndrome
C-	Filarisis can affect renal system causing renal system
D-	Filarisis can be diaqnosed by finding microfilaria in blood film
E-	Filarisis can be transmitted by female Culex, Aedes and Anopheles mosquito

13-	**All the following statements are true except?**
A-	Renal involvement in Schistosoma haematobium and S. mansoni occurs due to precipitation of immune complexes in the glomerular vascular bed
B-	Schistosoma haematobium can cause carcinoma of the bladder
C-	Both Schistosoma haematobium and Schistosoma mansoni can cause nephrotic syndrome
D-	Schistosoma mansoni can cause hydronephrosis
E-	Schistosoma haematobium can be diagnosed by finding egg in urine and Schistosoma mansoni can be diagnosed by finding egg in stool

14-	**Intermediate host of Schistosoma haematobium is……..?**		
A-	Bilunus trancatus	B-	Semisarcospira libertine
C-	Biomphalaria alexandrina	D-	None of the above

15-	**Infective stage of schistosoma haematobium is……..?**		
A-	Furcocercous cercaria	B-	Microcercous cercaria
C-	Encysted metacercaria	D-	None of the above

16-	**Katayama fever is serum sickness like syndrome?**		
A-	true	B-	False

17-	**Schistosoma haematobium is carcinogenic?**		
A-	True	B-	False

18-	**Intermediate host of Wuchreria bancrofti is.....?**		
A-	Male culex pipines	B-	Female culex pipines
C-	Filariform larva	D-	None of the above

19-	**To prevent hydatid disease we .....?**
A-	Avoid eating insufficiently cooked or washed foods
B-	Eradication of stray dogs
C-	Treatment of pet dogs
D-	All of the above

20-	**Culex transmits?**		
A-	Malaria	B-	Leismania
C-	Wuchereria bancrofti		

21-	**An Egyptian 12 years old child presented by frequent micturition, dysuria and terminal hematuria with mild anemia. Urine examination snowed egg with microscopic blood.** **The causative parasite is?**		
A-	Fasciolopsis Buski	B-	Schistosoma haematobium
C-	Schistosoma mansoni	D-	Schistosoma japonicum

22-	**Infective stage for this parasitic infection?**		
A-	Miracidium	B-	Egg
C-	Furcocercous Cercaria	D-	Encysted metacercaria

23-	**Intermediate host is?**		
A-	Fasciolopsis Buski	B-	Biomphalaria alexandrina
C-	Bulinus truncates	D-	Lymnaea truncatula

24-	**The main agent of pathology in schistosomiasis is?**		
A-	Eggs	B-	Adult worms
C-	Schistosomulum	D-	Furcocercus cercarae

25-	**The biomphelaria snail is the first intermediate host of?**		
A-	Fasciola spp.	B-	Schistosoma mansoni
C-	Heterophys heterophys	D-	Paragonimus westermani

26-	**The bolinus snail is the first intermediate host of?**		
A-	Fasciola spp.	B-	heterophys heterophys
C-	Paragonimus westermani	D-	Schistosoma haematubium

27-	**Mode of transmission of Schistosoma species is by?**
A-	Ingestion if encysted metacercariae found in undercocked fish
B-	Ingestion if encysted metacercariae found in undercocked crab
C-	Ingestion if encysted metacercariae on edible aquatic plants
D-	Penetration of Skin by furcocerus cercaria

28-	**The first Intermediate snail host of Schistosoma mansoni is?**		
A-	Semisulcospira	B-	Lymnea species
C-	Bulinus truncates	D-	Biomphelaria alexandrina

29-	**The First Intermediate snail host of Schistosoma haematobium is?**		
A-	Bulinus truncates	B-	Pirenella conica
C-	Semisulcospira	D-	Lymnea species

30-	**The type of cercaria in schistosoma spp is?**		
A-	Leptocercus cercaria	B-	Lophocercus cercaria
C-	Furcocercus cercaria	D-	Cysticecoid cercaria

31-	**The egg of Schistosoma mansoni have?**		
A-	Operculum	B-	Lateral spine
C-	Terminal spine	D-	Two mucoid plugs

32-	**The egg of Schistosoma haematobium have?**		
A-	Operculum	B-	Lateral spine
C-	Terminal spine	D-	Two mucoid plugs
33-	**Which one of the followings is wrong as regards the diagnosis of Schistosoma haematobium?**		
A-	Bladder biopsy	B-	Serological tests
C-	Detection of egg in stool	D-	Circumoval precipitin test

34-	**Which one of the followings is wrong as regards the diagnosis of Schistosoma mansoni?**		
A-	Rectal biopsy	B-	Serological tests
C-	Detection of ova in urine	D-	Circumoval precipitin test

﴿وَإِن تَعُدُّوا نِعْمَتَ اللَّهِ لَا تُحْصُوهَا إِنَّ الْإِنسَانَ لَظَلُومٌ كَفَّارٌ﴾

35-	**Which one of the followings is wrong as regards the types of Schistosomes ?**		
A-	Schistosoma mansoni	B-	Schistosoma haematobium
C-	Schistosoma braziliensis	D-	Schistosoma japonicum

36-	**In urinary bilharziasis terminal haematuria occurs in?**		
A-	Chronic stage	B-	Stage of Egg extrusion
C-	Stage of skin penetration	D-	Stage of adult migration

37-	**Bilharzia granuloma is due to?**		
A-	Toxin of parasite	B-	Pulmonary hypertension
C-	Presence of eggs in tissues	D-	Tissue damage and hemorrhage

38-	**Cancer bladder may occur as a complication of infection with?**		
A-	Fasciola gigantica	B-	Schistosom mansoni
C-	Paragonimus westermani	D-	Schistosoma haematobium

39-	**Schistosomulum is?**		
A-	Cercaria without tail	B-	Present inside the snail
C-	Cercaria with simple tail	D-	Emerged directly from egg

40-	**Schistosomulum mature into adult worm in?**		
A-	Systemic blood vessels	B-	Portal blood vessels
C-	Lymphatic vessels	D-	Biliary tracts

41-	**Which one of the following symptoms is specific for urinary bilharziasis is?**		
A-	Dysuria	B-	Chyluria
C-	Frank hematuria	D-	Terminal hematuria

42-	**Swimmer's itch occurs with the infection by?**		
A-	Fasciolia species	B-	Schistomsoma species
C-	Heterophys Heterophys	D-	Paragonimus westermani

43-	**Which stage of Schistosoma mansoni enters the bloodstream, finds its way to the hepatic portal veins and feeds on blood?**		
A-	Metacercaria	B-	Miracidium
C-	Sporocyst	D-	Cercaria

44-	**Which is wrong as regards the clinical presentation of Schistosomiasis haematobium?**		
A-	Terminal haematuria	B-	Obstructive uropathy
C-	Cancer bladder	D-	Dysentery

45-	**From the symptoms of Urinary Bilharziasis is?**		
A-	Dysuria	B-	Chyluria
C-	Frank hematuria	D-	Terminal hematuria

46-	**Which is true as regards the laboratory diagnosis of Schistosomiasis haematobium?**		
A-	Rectal biopsy	B-	Egg in stool
C-	Proctoscopy	D-	Cystoscopy

47-	**hydatid cyst can be found in several organs but are most frequent in the?**		
A-	Liver	B-	Lung
C-	Brain	D-	Bone

48-	**definitive host of Echinococcus granulosus is?**		
A-	Cattles	B-	Sheep
C-	Goats	D-	Dogs

49-	**Hydatid disease is frequently associated with?**		
A-	Basophilia	B-	Hemophilia
C-	Eosinophilia	D-	Neutrophilia

50-	**In hydatid disease man act as?**		
A-	Definitive host	B-	Reservoir host
C-	Accidental host	D-	Intermediate host

51-	**Hydatid disease is caused by?**		
A-	Echinococcus granulosus	B-	Enterobius vermicularis
C-	Trichuris trichiura	D-	Taenia solium

52-	**Which one of the following is wrong as regards the complications of lymphatic obstruction of Wuchereria bancrofti?**		
A-	Elephantiasis	B-	Haematuria
C-	Hydrocele	D-	Chyluria

53-	**Arthropod vector of Wuchereria bancrofti is?**		
A-	Mansonia	B-	Black flies
C-	Female culex spp.	D-	Female anopheles spp

54-	**As regards Leishmania donovani causes?**		
A-	Tropical pulmonary eosinophilia	B-	Visceral larva migrans
C-	Chagas disease	D-	Kala-azar

55-	**Microfilariae of Wuchereria bancrofti are?**
A-	Sheathed with 2 widely spaced nuclei & blunt tip at tail end
B-	Sheathed microfilaria with nuclei up to rounded tail tip
C-	Sheathed microfilaria with no nuclei at tail tip
D-	Unsheathed microfilaria with no nuclei

56-	**As regards Leishmania species?**
A-	They are extracellular parasite
B-	They have fecal-oral transmission
C-	They have developing stages inside the red blood cells
D-	They are multiplying in the cells of reticuloendothelian system

57-	**Nephrotic syndrome occurs in chronic infection with?**		
A-	Plasmodium ovale	B-	Plasmodium vivax
C-	Plasmodium malariae	D-	Plasmodium falciparum

58-	**Infection of man is infected with babesia through the bite of?**		
A-	Soft tick	B-	Hard tick
C-	Female Sandy fly	D-	Female Anopheline mosquitoes

59-	**Which one of the following statements is true as regards plasmodium falciparum infection?**
A-	Relapses occur every 3 months
B-	It causes bengin tertian infection
C-	It is characterized by severe anemia
D-	Characteristically paroxysms occurs every 72 hours

60-	**Which one of the following infections is not associated with splenomegaly is not caused by?**		
A-	Schistomiasis	B-	Ascariasis
C-	Kala-azar	D-	Malaria

61-	**As regards cutaneous Leishmaniasis it is?**		
A-	Diagnosed by bone marrow smear	B-	Caused by leishmania donovani
C-	Diagnosed by xenodiagnosis	D-	Transmitted by sand fly

62-	**As regards malaria infection asexual phase takes place in the?**		
A-	Red blood cells	B-	Macrophage
C-	Mosquitoes	D-	Liver cells
E-	A&D		

63-	As regards malaria infection sexual phase takes place in the?		
A-	Red blood cells	B-	Macrophage
C-	Mosquitoes	D-	Liver cells

64-	As regards Malaria attacks (paroxysms) are due to?
A-	Rupture of Hepatocytes with release Merozoites
B-	Rupture of RBCs with release of Merozoites
C-	Invasion of Hepatocytes by Merozoites
D-	Invasion of RBCs by Merozoites

65-	Which one of the followings does leishmania amastigote inhabit?		
A-	Hepatocyte	B-	Erythrocytes
C-	Macrophages	D-	Squamous cells

66-	Leishmaniasis is transmitted by?		
A-	Female Anopheles	B-	Triatoma megesta
C-	Sandy fly	D-	Tse tse fly

ANSWERS - PARASITOLOGY									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
A	C	D	A	B	D	D	A	A	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	C	D	A	A	A	A	B	D	C
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	C	C	A	B	D	D	D	A	C
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
B	C	C	C	C	B	C	D	A	B
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
D	B	D	D	D	D	A	D	C	D
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
A	B	C	D	C	D	C	B	D	D
61	**62**	**63**	**64**	**65**	**66**	**67**	**68**	**69**	**70**
D	E	C	B	C	C	-	-	-	-

Pathology



❖ Congenital anomalies of the kidney

1-	**In the usual adult type of polycystic kidneys, findings often include?**		
A-	Involvement of both kidneys	B-	Another family member with the disease
C-	Enlarged kidneys	D-	Renal failure late in course
E-	All of the above		

2-	**In adult polycystic disease ,all of the following statements is true except?**
A-	The disease is familial
B-	The renal involvement is occasionally unilateral
C-	The disease is generally asymptomatic until early adult or middle life
D-	Hematuria, hypertension and palpable abdominal masses are frequent clinical findings in symptomatic patients

3-	**Which of the following are true regarding adult polycystic disease of the kidney?**
A-	The disease is generally asymptomatic until early adult and middle life
B-	Hematuria, hypertension, and palpable abdominal masses are frequent clinical findings in symptomatic patients
C-	Both
D-	Neither

4-	**Which of the following is a typical features of adult polycystic kidney disease include?**		
A-	Involvement of both kidneys	B-	Autosomal dominant inheritance
C-	Enlarged kidneys	D-	Cysts in the liver
E-	All of the above		

5-	**Polycystic disease of the kidney is characteristically associated with?**		
A-	Amyloidosis	B-	The nephrotic syndrome
C-	Both	D-	Neither

6-	**Infantile polycystic kidney disease has the following features EXCEPT?**
A-	It has autosomal dominant inheritance
B-	It is invariably bilateral
C-	It often manifests in adults
D-	The condition is frequently associated with multiple hepatic cysts

7-	**The adult polycystic kidney disease is characterized by?**		
A-	Involves both kidneys	B-	Massive enlargement of the kidneys
C-	Autosomal dominant	D-	All of the above

8-	**Infantile polycystic kidney disease has the following features EXCEPT?**
A-	It has autosomal recessive inheritance
B-	It is invariably bilateral
C-	The condition often manifests in adults
D-	The condition is frequently associated with multiple hepatic cysts

9-	**Which of the following diseases is characterized by autosomal recessive inheritance?**		
A-	Adulthood polycystic kidney disease	B-	Childhood polycystic kidney disease
C-	Familial polyposis	D-	Neurofibromatosis

10-	**Horse-shoe kidney is a?**		
A-	Senile kidney	B-	Congenitally malformed kidney
C-	Hypertensive kidney	D-	Atherosclerotic kidney
E-	Ectopic kidney		

11-	**Adult polycystic kidney is characterized by all EXCEPT?**
A-	Inherited as an autosomal dominant trait
B-	Commonly leads to renal failure
C-	Bilateral
D-	Shows multiple cysts communicating with the pelvis
E-	The cysts are lined by cuboidal or columnar epithelium

ANSWERS – Congenital anomalies of the kidney									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
E	B	C	E	D	A	D	C	B	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	-	-	-	-	-	-	-	-	-

يقول ابن القيم – رحمه الله –:

«إذا أَحَبَّ اللهُ عبدًا اصطَنَعَه لنَفسِه، واجتَباهُ لمَحبَّتِه، واستَخلَصَهُ لعبادته، فشَغَلَ هَمَّهُ به ولسانه بذِكره، وجَوارحه بخِدمته» .

❖ Glomerulonephritis

1-	All of the following features are characteristic of the nephrotic syndrome except?		
A-	marked proteinuria	B-	hypoalbuminemia
C-	edema	D-	hypertension

2-	Hematuria is least likely to occur in?
A-	papillary transitional cell carcinoma of renal pelvis, ureter, or bladder
B-	renal tubular adenocarcinoma
C-	acute post-infectious glomerulonephritis
D-	urinary lithiasis
E-	minimal change disease

3-	Urinalysis results include proteinuria, many red blood cells, and red blood cell casts, and 1-2 white blood cells per high power field Which of the following diseases best fits with these findings?		
A-	acute cystitis	B-	acute pyelonephritis
C-	chronic pyelonephritis	D-	acute post-streptococcal glomerulonephritis
E-	minimal change disease		

4-	All of the following is true of acute poststreptococcal glomerulonephritis except?
A-	becomes rapidly progressive glomerulonephritis in 50% of children
B-	is a proliferative glomerulonephritis
C-	preceded by infective episode
D-	shows dome-shaped subepithelial humps
E-	usually affects children

5-	The majority of patients with acute post-streptococcal glomerulonephritis?
A-	die from congestive heart failure in a few weeks
B-	die from renal failure in a few weeks
C-	pass through a latent period but eventually develop chronic glomerulonephritis
D-	progress through subacute phase to chronic glomerulonephritis
E-	recover completely

6-	A 12-year-old boy with septicemia develops hematuria, hypertension, and oliguria. Microscopic examination reveals enlarged glomeruli with diffuse mesangial and endothelial hypercellularity. The best diagnosis is?		
A-	acute cortical necrosis	B-	acute proliferative glomerulonephritis
C-	membranoproliferative glomerulonephritis	D-	rapidly progressive glomerulonephritis

7-	**A patient with hemoptysis and renal failure has a renal biopsy which reveals crescentic proliferative glomerulonephritis. Immunofluorescence reveals linear deposits of IgG and C3.The diagnosis is?**		
A-	acute post-infectious glomerulonephritis	B-	membranous nephropathy
C-	Rapid progressive	D-	minimal change disease

8-	**The most characteristic feature of rapidly progressive glomerulonephritis is?**		
A-	crescent formation	B-	hyaline nodules
C-	sub-epithelial dense deposits	D-	thickened capillary loops
E-	wire-loop lesions		

9-	**The nephrotic syndrome is characterized by all of the following except?**		
A-	generalized edema	B-	hematuria
C-	proteinuria	D-	hypoproteinemia
E-	hyperlipidemia		

10-	**Minimal change disease and membranous glomerulonephritis share which of the following characteristics?**		
A-	nephrotic syndrome	B-	IgG deposits in glomeruli
C-	good prognosis with steroid therapy	D-	capillary loop thickening by light microscopy

11-	**Deposition of IgG on or in the glomerular basement membrane can be seen in all of the following except?**		
A-	acute post-streptococcal glomerulonephritis	B-	minimal change glomerulonephritis
C-	lupus erythematosis glomerulonephritis	D-	Goodpasture's syndrome
E-	membranous nephropathy		

12-	**Nephrotic syndrome is characterized by?**		
A-	hematuria, proteinuria, hypertension	B-	pyuria, oliguria, hematuria
C-	hematuria, azotemia, hypertension	D-	proteinuria, edema, hyperlipidemia
E-	bacteriuria, azotemia, hypertension		

13-	**In all of the following diseases, glomerular damage is due to deposition of circulating immune complexes except for?**
A-	systemic lupus erythematosis
B-	membranous glomerulonephritis(MGN)
C-	acute post-streptococcal glomerulonephritis(PSGN)
D-	minimal change disease

14-	**A 50-year-old male develops acute renal failure with dark urine one week after a flulike illness. A renal biopsy is likely to reveal?**		
A-	renal infarction	B-	crescentic glomerulonephritis
C-	myeloma kidney	D-	amyloidosis
E-	acute tubular necrosis		

15-	**Most children with poststreptococcal glomerulonephritis?**		
A-	develop nephrotic syndrome	B-	progress to chronic glomerulonephritis
C-	develop hypertensive cardiovascular disease	D-	recover spontaneously
E-	have B strep bacteria in their urine		

16-	**The nephrotic syndrome is primarily due to?**
A-	diabetic glomerulosclerosis
B-	toxic necrosis of renal tubular epithelium
C-	increased permeability of the glomerular filtration barrier
D-	narrowing of the afferent arteriole

17-	**The most common appearance of the glomeruli in adults with the nephrotic syndrome is?**
A-	proliferative glomerulonephritis
B-	membranous glomerulonephritis
C-	rapidly progressive "crescentic" glomerulonephritis
D-	normal appearing glomeruli
E-	wire loops

18-	**Which is the most characteristic feature in the urine sediment of patients with poststreptococcal glomerulonephritis?**		
A-	epithelial casts	B-	white cell casts
C-	red blood cells and casts	D-	marked proteinuria

19-	**Majority of cases of membranous GN have the following etiology?**		
A-	SLE	B-	Viral infection
C-	History of drug	D-	idiopathic

20-	**Post-streptococcal glomerulonephritis associated with?**
A-	Subepithelial deposition of immune complexes
B-	Nephrotic syndrome is a common syndrome
C-	Spontaneous resolution in most adults cases
D-	Streptococci can be cultured from the urine

21-	**In minimal change disease of the glomeruli?**
A-	The incidence is greatest between the ages of 20 and 40 years
B-	Fusion of the foot processes of the podocytes is seen on electron microscope
C-	A non-selective proteinuria is usual
D-	Light microscopy shows crescent formation in glomeruli

22-	**The nephrotic syndrome, which of the following is NOT a feature?**
A-	May be caused by renal amyloidosis
B-	Is associated with hypolipidemia
C-	In children, is usually due to minimal change glomerulonephritis
D-	Is characterized by proteinuria, hypoalbuminemia and edema

23-	**Hematuria, hypertension, edema and oliguria are features of?**		
A-	Nephrotic syndrome	B-	Nephritic syndrome
C-	Both	D-	None

24-	**In antibody-mediated glomerular injury, Cytotoxic antibodies are directed against the following EXCEPT?**		
A-	Mesangial cells	B-	Endothelial cells
C-	Visceral epithelial cell glycoprotein	D-	Tubular epithelium

25-	**What finding on microscopic urinalysis indicates that hematuria in a patient is caused by a renal process, rather than bleeding from another site in the urinary tract?**		
A-	Blood clots	B-	Hemoglobin crystals
C-	Phagocytosed hemoglobin	D-	Red blood cell casts
E-	White blood cell casts		

26-	**The most common features of nephritic syndrome include the following EXCEPT?**		
A-	Heavy proteinuria	B-	Hypertension
C-	Microscopic hematuria	D-	Oliguria

27-	**The most common form of glomerulonephritis (GN) in adults is?**		
A-	Minimal change disease	B-	Membranous GN
C-	Membranoproliferative GN	D-	Focal segmental GN
E-	Crescentic GN		

وما كلُّ برقٍ لاحَ لي يستفزُّني *** ولا كلُّ من في الأرضِ أرضاه منعَّما

28-	**The most frequent form of primary glomerular disease in children is?**		
A-	Minimal change disease	B-	Acute glomerulonephritis
C-	Membranous GN	D-	Membranoproliferative GN

29-	**A concerned mother brings her 6-year-old son to the pediatrician because he has developed a "puffy face."On examination, the child has a blood pressure of 90/60 mm Hg,marked peri-orbital edema, and pitting edema of both the hands and feet. Cardiac examination is unremarkable, and he has no splenomegaly or signs of liver disease. Laboratory values are notable for decreased serum albumin and increased total cholesterol. A urinalysis reveals 4+ proteinuria but no red blood cells, white blood cells, or casts. Which of the following is the most likely diagnosis?**		
A-	Good pasture syndrome	B-	Minimal change nephropathy
C-	post streptococcal glomerulonephritis	D-	Systemic lupus erythematosus
E-	Wegener granulomatosis		

30-	**A 3-year-old boy is brought to the physician by his mother because of a 2-day history of "puffy eyes and smoky urine: 'His mother says that he had a "skin infection"2 weeks before the onset of these symptoms. His temperature is 38.1° and his blood pressure is 140/90mm Hg.He has significant peri-orbital edema. Laboratory studies show elevated levels of antistreptolysin a titers and a low serum complement. Which of the following is the most likely electron microscopic finding associated with this patient condition?**
A-	Glomerular basement membrane disruption and no deposits
B-	Effacement of epithelial foot processes
C-	Mesangial immune complex deposits
D-	Sub-endothelial and mesangial immune complex deposits
E-	Sub-epithelial immune complex deposits

31-	**The following statements are true about post-streptococcal glomerulonephritis except?**
A-	It is an immune complex disease
B-	Acute renal failure is the most common presenting feature
C-	The glomeruli appear microscopically hypercellular
D-	It mostly affects children & young adults

32-	**Which of the following statements is true about crescentic glomerulonephritis?**
A-	It is characterized by glomerular necrosis
B-	The bowman's capsules show parietal crescents
C-	The disease often rapidly progresses to renal failure
D-	All of the above

33-	Which of the following diseases is most liable to rapid progression to uremia?		
A-	Acute post-streptococcal glomerulonephritis	B-	Crescentic glomerulonephritis
C-	Benign hypertension	D-	Bilharzial stricture of one ureter

34-	Acute diffuse proliferative glomerulonephritis follows infection by?		
A-	Streptococcus viridians	B-	Streptococcus haemolyticus
C-	Streptococcus faecalis	D-	Staphylococcus aureus
E-	E.coli		

35-	Acute diffuse proliferative glomerulonephritis is NOT characterized by?		
A-	Hypercellular glomeruli	B-	Infiltration of the glomeruli by polymorphs
C-	Proliferation of endothelial cells	D-	Proliferation of mesangial cells
E-	Epithelial crescents in the glomeruli		

36-	Acute diffuse proliferative glomerulonephritis is due to?
A-	Blood-borne infection
B-	Ascending infection
C-	Deposition of immune complexes in the glomeruli
D-	Autoimmune disease
E-	None of the above

37-	The diagnostic histologic feature of rapidly progressive glomerulonephritis is?		
A-	Hypercellular glomeruli	B-	Segmental necrosis of the glomeruli
C-	Epithelial Crescents	D-	Protein casts in the tubules
E-	Thick-walled blood vessels		

38-	The etiology of rapidly progressive glomerulonephritis may be?
A-	Post-streptococcal
B-	Systemic lupus erythematosus
C-	Formation of antiglomerular basement membrane antibody
D-	All of the above

39-	Nephritic syndrome is characterized by all EXCEPT?		
A-	Hematuria	B-	Heavy proteinuria
C-	Oliguria	D-	Hypertension

40-	Nephrotic syndrome is NOT characterized by?		
A-	Heavy proteinuria	B-	Hypoproteinemia
C-	Oedema of the face only	D-	Hyperlipidemia

41-	**Oedema of the nephritic syndrome appears in?**		
A-	Face	B-	Lower limbs
C-	Back	D-	All over the body

42-	**Oedema in the nephritic syndrome is due to?**
A-	Hypoproteinemia
B-	Excessive reabsorption of water and salt by the tubules
C-	Increased hydrostatic pressure of the blood
D-	All of the above
E-	None of the above

43-	**Oliguria in the nephritic syndrome is due to?**		
A-	Decreased blood volume	B-	Decreased renal blood flow
C-	Decreased water intake	D-	All of the above
E-	None of the above		

44-	**Hypoproteinemia in the nephrotic syndrome is due to?**
A-	Loss of proteins in urine
B-	Decreased protein intake
C-	Decreased absorption of proteins in the intestine
D-	Escape of proteins to the interstitial tissue
E-	None of the above

45-	**The following is NOT a cause of nephrotic syndrome?**		
A-	Acute diffuse proliferative glomerulonephritis	B-	Membranous glomerulonephritis
C-	Membrano-proliferative glomerulonephritis	D-	Minimal change glomerulonephritis
E-	Systemic lupus erythematosus		

46-	**Epithelial crescents are characteristic of?**		
A-	Acute diffuse proliferative glomerulonephritis	B-	Rapidly progressive glomerulonephritis
C-	Membranous glomerulonephritis	D-	Membrano-proliferative glomerulonephritis
E-	Minimal change glomerulonephritis		

47-	**The following changes occur in minimal change glomerulonephritis?**		
A-	Proliferation of the mesangial cells	B-	Proliferation of the capsular epithelium
C-	All of the above	D-	None of the above

48-	**The following disease is diagnosed only by electron microscopy?**		
A-	Acute diffuse proliferative glomerulonephritis	B-	Rapidly progressive glomerulonephritis
C-	Membranous glomerulonephritis	D-	Membrano-proliferative glomerulonephritis
E-	Minimal change glomerulonephritis		

49-	**Subepithelial electron dense deposits are characteristic of?**		
A-	Acute diffuse proliferative glomerulonephritis	B-	Rapidly progressive glomerulonephritis
C-	Membranous glomerulonephritis	D-	Membrano-proliferative glomerulonephritis
E-	Minimal change glomerulonephritis		

50-	**Fusion of the foot processes of the podocytes is characteristic of ?**		
A-	Acute diffuse proliferative glomerulonephritis	B-	Rapidly progressive glomerulonephritis
C-	Membranous glomerulonephritis	D-	Membrano-proliferative glomerulonephritis
E-	Minimal change glomerulonephritis		

51-	**The following is NOT a cause of hematuria?**		
A-	Schistosomiasis of the bladder	B-	Renal stones
C-	Nephrotic syndrome	D-	Tumors of the urinary system
E-	Congenital polycystic kidney		

52-	**A woman suffering from systemic lupus erythematosus is now complaining of hematuria, renal biopsy showed generalized glomerular hypercellularity, thickening of the glomerular capillary walls and epithelial crescents. The most possible diagnosis is?**		
A-	Focal glomerulonephritis	B-	Membranous
C-	Rapidly progressive glomerulonephritis	D-	Chronic glomerulonephritis
E-	Membranoproliferative glomerulonephritis		

53-	**A child suffered from upper respiratory tract infection. Two weeks later, he presented with loin pain, edema of the face with puffiness of the eyelids. His renal biopsy will show the picture of?**		
A-	Acute diffuse proliferative glomerulonephritis	B-	Membranous glomerulonephritis
C-	Membranoproliferative glomerulonephritis	D-	Chronic glomerulonephritis
E-	Focal glomerulonephritis		

54-	**A child suffering from post-streptococcal glomerulonephritis may die due to?**		
A-	Acute renal failure	B-	Complications of hypertension
C-	chronic renal failure	D-	Any of the above
E-	None of the above		

55-	Upper respiratory tract streptococcal infection may be followed by?		
A-	Rheumatic fever	B-	Acute diffuse proliferative glomerulonephritis
C-	Rapidly progressive glomerulonephritis	D-	Any of the above
E-	None of the above		

56-	The diagnostic histologic feature of rapidly progressive glomerulonephritis is?		
A-	Hypercellular glomeruli	B-	Segmental necrosis of the glomeruli
C-	Epithelial crescents	D-	Protein casts in the tubules

57-	All of the following cause Nephrotic syndrome except?		
A-	Minimal change disease	B-	Membranoproliferative Glomerulonephritis
C-	Acute diffuse proliferative Glomerulonephritis	D-	Membranous Glomerulonephritis

ANSWERS – Glomerulonephritis									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	E	B	A	E	B	C	A	B	A
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	D	D	B	D	C	B	C	D	A
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
B	B	B	D	D	A	B	A	B	D
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
B	B	B	B	E	C	C	D	B	C
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
A	B	B	A	A	B	D	E	C	E
51	**52**	**53**	**54**	**55**	**56**	**57**	**58**	**59**	**60**
C	C	A	A	D	C	C	-	-	-

قال ابن تيمية – رحمه الله –:

" ومن أصغى إلى كلام الله وكلام رسوله صلى الله عليه وسلم بعقله وتدبره بقلبه وجد فيه من الفهم والحلاوة والهُدى وشفاء القلوب والبركة والمنفعة ما لا يجده في شيء من الكلام لا نظمًا ولا نثرًا "

❖ Bladder & stones

1-	**An irritable female infant, age 2 years, has a fever of 104° F and bloody urine. Microscopic examination of the urine reveals red blood cells and neutrophils but no casts. The most likely diagnosis is?**		
A-	sarcoma botryoides	B-	acute cystitis
C-	urinary stone	D-	acute glomerulonephritis

2-	**Schistosomiasis of the bladder predisposes to?**		
A-	Cystitis	B-	Stone formation
C-	Bladder carcinoma	D-	All of the above
E-	None of the above		

3-	**The following is NOT a metaplastic lesion?**		
A-	Brunn's nests	B-	Cystitis glandularis
C-	Leukoplakia	D-	Nephrogenic adenoma

4-	**The metaplastic lesions which may occur in the bladder epithelium include?**		
A-	Squamous metaplasia without keratinization	B-	Squamous metaplasia with keratinization
C-	Glandular metaplasia	D-	Mesonephric metaplasia
E-	All of the above		

5-	**Bilharziasis may lead to all EXCEPT?**		
A-	Portal hypertension	B-	Stone formation
C-	Carcinoma of urinary	D-	Carcinoma of the colon
E-	Urinary tract infection		

6-	**Urinary calculi are characterized by?**
A-	May cause glomerulonephritis
B-	Are more common in temperate than tropical climate
C-	May induce squamous metaplasia in surrounding urothelium
D-	Are most commonly made of uric acid
E-	Are rarely made of calcium oxalate compounds

7-	**The crystalloids that are the major component of renal calculi are calcium plus?**		
A-	Cysteine and oxalate	B-	Cysteine and phosphate
C-	Oxalate and phosphate	D-	Oxalate and urate

﴿ إِنِ الْحُكْمُ إِلَّا لِلَّهِ عَلَيْهِ تَوَكَّلْتُ وَعَلَيْهِ فَلْيَتَوَكَّلِ الْمُتَوَكِّلُونَ ﴾

8-	**The following type of renal calculi are radiolucent?**		
A-	Calcium oxalate	B-	Stag horn stone
C-	Uric acid	D-	Calcium phosphate

9-	**Any of the following may be a cause of urinary stones EXCEPT?**		
A-	Excessive intake of water	B-	Hyperparathyroidism
C-	Gout	D-	Cystitis

10-	**The following is NOT a cause of renal calculi?**		
A-	Vitamin A deficiency	B-	Vitamin D deficiency
C-	Urinary tract infection	D-	Urinary tract obstruction
E-	Hyperparathyroidism		

11-	**The following is NOT a type of renal stones?**		
A-	Oxalate stones	B-	Urate stones
C-	cholesterol stones	D-	Phosphate stones
E-	Cystine stones		

12-	**The following type of renal stones usually occurs on top of infection?**		
A-	Oxalate stones	B-	Urate stones
C-	Phosphate stones	D-	Cystine stones

13-	**The following type of renal stones occurs in the urinary bladder?**		
A-	Oxalate stones	B-	Urate stones
C-	Phosphate stones	D-	Cystine stones

14-	**The following type of renal stones usually has a stag horn appearance?**		
A-	Oxalate stones	B-	Urate stones
C-	Phosphate stones	D-	Cystine stones

15-	**The following type of renal stones is multiple and spiky?**		
A-	Oxalate stones	B-	Phosphate stones
C-	Urate stones	D-	Cystine stones

16-	**The following type of renal stones is commonly associated with hematuria?**		
A-	Oxalate stones	B-	Phosphate stones
C-	Urate stones	D-	Cystine stones

17-	**The following type of renal stones is very rare?**		
A-	Oxalate stones	B-	Urate stones
C-	Phosphate stones	D-	Cystine stones

ANSWERS – Bladder & stones									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	D	A	E	D	C	C	C	A	B
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
C	C	C	B	A	A	D	-	-	-

مِن عُقُوباتِ الذُّنُوبِ: أَنَّها تُزِيلُ النِّعَمَ، وتُحِلُّ النِّقَمَ، فَما زالَتْ عَنِ العَبْدِ نِعْمَةٌ إلّا بِذَنْبٍ، ولا حَلَّتْ بِهِ نِقْمَةٌ إلّا بِذَنْبٍ، كَما قالَ عَلِيُّ بْنُ أبِي طالِبٍ -رَضِيَ اللَّهُ عَنْهُ-: ما نَزَلَ بَلاءٌ إلّا بِذَنْبٍ، ولا رُفِعَ إلّا بِتَوْبَةٍ.

وَقَدْ قالَ تَعالى: ﴿وَما أَصابَكُمْ مِن مُصِيبَةٍ فَبِما كَسَبَتْ أَيْدِيكُمْ وَيَعْفُو عَنْ كَثِيرٍ﴾

❖ Tubulo-interstitial diseases

1-	**A 22-year-old pregnant female has flank pain and fever. Her urine contains white blood cell casts. It can be safely assumed that she has at least?**		
A-	chronic glomerulonephritis	B-	analgesic nephritis
C-	acute pyelonephritis	D-	ectopic pregnancy
E-	pre-eclampsia(toxemia of pregnancy)		

2-	**The most common organism associated with acute pyelonephritis is?**		
A-	Staphylococcus aureus	B-	chlamydia
C-	beta-hemolytic streptococci	D-	Campylobacter
E-	E.coli		

3-	**The most common mechanism in pathogenesis of chronic pyelonephritis?**		
A-	Ascending infection	B-	Reflux nephropathy
C-	Obstructive nephropathy	D-	Hematogenous infection

4-	**The following factors are predisposing factor to pyelonephritis EXCEPT?**		
A-	Diabetes mellitus	B-	Pregnancy
C-	Vesico-ureteric reflux	D-	Males less than 40 years old
E-	Cystoscopy		

5-	**All of the followings are associated with acute pyelonephritis, EXCEPT?**
A-	Unilateral or bilateral suppuration
B-	Acute tubular inflammation and destruction
C-	Glomerular destruction is a common feature
D-	Due to hematogenous or ascending spread

6-	**In ischemic acute tubular necrosis, the following holds true EXCEPT?**
A-	There is dilatation of proximal and distal convoluted tubules
B-	There is disruption of tubular basement membrane
C-	There is diffuse tubular necrosis
D-	Tubular lumina contain casts

7-	**A severely ill patient in shock develops acute renal failure with oliguria and azotemia. The urine osmolality approaches that of the glomerular ultrafiltrate (specific gravity 1.010).A renal biopsy would most likely show which of the following?**		
A-	Acute pyelonephritis	B-	Acute tubular necrosis
C-	Chronic pyelonephritis	D-	Crescentic glomerulonephritis
E-	Renal cell carcinoma		

8-	**Renal diseases producing systemic hypertension includes all of the following EXCEPT?**		
A-	Acute glomerulonephritis	B-	Acute pyelonephritis
C-	Chronic glomerulonephritis	D-	Chronic pyelonephritis
E-	Renal vasculitis		

9-	**Which of the following pathological change is most likely to be seen in acute pyelonephritis?**		
A-	Cortical atrophy	B-	Papillary necrosis
C-	Polycystic change	D-	Renal cell carcinoma
E-	Antigen-antibody complexes		

10-	**Which of the following is characteristic for chronic pyelonephritis?**		
A-	Chronic inflammatory cellular infiltrate	B-	Renal tubular casts
C-	Cortico-medullary scarring	D-	Tubular epithelial atrophy
E-	All of the above		

11-	**White blood cell casts are usually seen in?**		
A-	Chronic pyelonephritis	B-	Acute pyelonephritis
C-	Acute glomerulonephritis	D-	Chronic glomerulonephritis
E-	All of the above		

12-	**The most common mechanism in pathogenesis of chronic pyelonephritis is?**		
A-	Ascending infection	B-	Reflux nephropathy
C-	Obstructive nephropathy	D-	Hematogenous infection
E-	All of the above		

13-	**Chronic pyelonephritis is characterized by the following features EXCEPT?**
A-	The kidneys are hugely enlarged and show pelvicalyceal dilatation
B-	Females are more commonly affected than females
C-	Urinary tract obstruction by stones is a commonly associated condition
D-	The tubules often show thyroid-like casts

14-	**Which of the following diseases is associated with pyuria (pus in urine)?**		
A-	Acute post-streptococcal glomerulonephritis	B-	Acute pyelonephritis
C-	Hydronephrosis	D-	Chronic glomerulonephritis

15-	**Factors predisposing to pyelonephritis include all EXCEPT?**		
A-	Urinary tract obstruction	B-	Urinary tract instrumentation
C-	Schistosomiasis	D-	Hypertension
E-	Diabetes		

16-	**Females are more susceptible to urinary tract infection because?**		
A-	The immune system is weaker in females	B-	High estrogen level
C-	Low testosterone level	D-	short urethra
E-	Higher incidence of diabetes		

17-	**Routes of infection in pyelonephritis?**		
A-	Ascending infection from the urinary bladder	B-	Lymphatic spread from the intestine
C-	Blood spread from a septic focus	D-	All of the above
E-	None of the above		

18-	**Thyroidisation is a histologic feature of?**		
A-	Hydronephrosis	B-	Pyonephrosis
C-	Acute pyelonephritis	D-	Chronic pyelonephritis
E-	All of the above		

19-	**A 40-year-old woman presented with chronic renal failure-pyelography showed dilation of the pelvicalyceal system. Renal biopsy revealed tubular atrophy, periglomerular fibrosis and interstitial inflammatory cells. The most possible diagnosis is?**		
A-	Renal stones	B-	Chronic glomerulonephritis
C-	Chronic pyelonephritis	D-	Membranous glomerulonephritis
E-	Diffuse proliferative glomerulonephritis		

ANSWERS – Tubulo-interstitial diseases									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	E	B	D	C	C	B	B	B	E
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
B	B	A	B	D	D	D	D	C	-

قال قتادة – رحمه الله –:

إن هذا القرآن يدلكم على دائكم ودوائكم؛ فأما داؤكم فالذنوب والخطايا، وأما دواؤكم فالاستغفار.

❖ Tumors

1-	**A 48-year-old male has hemoptysis, microscopic hematuria, and a mass at the upper pole of the right kidney. The most likely diagnosis is?**		
A-	Goodpasture's syndrome	B-	Cushing's syndrome
C-	primary aldosteronism	D-	renal adenocarcinoma
E-	hepatocellular carcinoma		

2-	**Schistosomiasis of the urinary bladder is implicated in the following type of bladder tumor?**		
A-	Transitional cell carcinoma	B-	Squamous cell carcinoma
C-	Adenocarcinoma	D-	Adenoacanthoma

3-	**Renal cell carcinoma (RCC)characterized by?**
A-	Papillary type is hypervascular
B-	Clear cell variant is most common
C-	Chromophobe RCC shows perivascular arrangements of malignant cells
D-	All of the above
E-	None of the above

4-	**All are characters of renal carcinoma, EXCEPT?**
A-	Early invades the renal vein
B-	Is more frequent in children than in adults
C-	Is often composed of large cells with abundant clear cytoplasm
D-	May present with total and profuse hematuria
E-	Is one of the tumors which often metastasize to the bone

5-	**Wilm's tumor, characterized by the followings EXCEPT?**
A-	Involves both kidneys in about 7% of cases
B-	Usually presents as an abdominal mass
C-	Rarely metastasizes to the lung
D-	Usually affects infants

6-	**Which of the following are true for bladder carcinoma?**
A-	It is common in lateral and posterior wall
B-	Ulcerative carcinoma is the commonest gross pattern
C-	Transitional cell carcinoma is common in bilharzial cases
D-	Spread to hypogastric and iliac lymph node is common in bilharziasis

7-	**Painless hematuria in a heavy smoker old man is probably due to?**		
A-	Acute cystitis	B-	Acute pyelonephritis
C-	Bladder calculi	D-	Carcinoma of the bladder
E-	Prostatic carcinoma		

8-	**Gross features of renal cell carcinoma include all of the following EXCEPT?**
A-	It is commonly centered in the cortex
B-	It is surrounded by true capsule
C-	It is commonly well-circumscribed
D-	Cut section is golden yellow
E-	Cut section shows areas of hemorrhage and necrosis

9-	**The commonest urinary tract malignant tumor in children is?**		
A-	Transitional cell carcinoma	B-	Leiomyosarcoma
C-	Rhabdomyosarcoma	D-	Wilm's tumor
E-	Liposarcoma		

10-	**Which of the following is true about nephroblastoma (Wilm's tumor)?**
A-	It is a malignant embryonic tumor that may sometimes affect both kidneys
B-	It most commonly affects children under the age of 15 years
C-	It consists of a variable mixture of blastemal, epithelial, and mesenchymal structures
D-	All of the above

11-	**Which of the following is true about renal cell carcinoma?**
A-	The tumor usually arises at one pole of the kidney
B-	The clear cell type is the most common microscopic type
C-	It may lead to pulmonary cannon ball metastases
D-	All of the above

12-	**Histologic features of renal cell carcinoma include all EXCEPT?**
A-	It may show tubular or papillary formations
B-	The cells have clear or granular cytoplasm
C-	The cells contain mucin
D-	The cell may show sarcomatoid appearance
E-	The stroma is very vascular

13-	**The most important prognostic factor in renal cell carcinoma is?**		
A-	The patient's age	B-	The tumor size
C-	The degree of nuclear atypia	D-	The degree of papillary formation
E-	The amount of intracellular glycogen		

14-	**Nephroblastoma is called?**		
A-	Warthin's tumor	B-	Wilm's tumor
C-	Benign mixed tumor	D-	Pancoast's tumor
E-	Yolk sac tumor		

15-	**Wilm's tumor consists of?**		
A-	Blastematous tissue	B-	Mesenchymal tissue
C-	Epithelial tissue	D-	All of the above
E-	None of the above		

16-	**The blastomatous tissue consists of?**		
A-	Spindle-shaped cells	B-	Small primitive cells
C-	Small muscle cells	D-	Malignant epithelial cells
E-	Neuroendocrine cells		

17-	**Leukoplakia of the urinary bladder means transformation of the bladder epithelium into?**		
A-	Non-keratinized squamous epithelium	B-	Keratinized epithelium
C-	Glandular epithelium	D-	Atrophic epithelium
E-	Dysplastic epithelium		

18-	**The International WHO histologic grading of transitional cell carcinoma depends on all EXCEPT?**		
A-	Degree of papillary formation	B-	Degree of squamous differentiation
C-	Degree of cytologic atypia	D-	Degree of mitotic activity

19-	**The most prognostic factor in bladder carcinoma is?**		
A-	The patient's age	B-	The tumor size
C-	The depth of tumor invasion	D-	The histologic grade
E-	The histologic type		

20-	**Histologic types of bladder carcinoma include?**		
A-	Squamous cell carcinoma	B-	Transitional cell carcinoma
C-	Adenocarcinoma	D-	Sarcomatoid carcinoma
E-	All of the above		

ANSWERS – Tumors									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	B	D	B	C	A	D	B	D	D
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	E	C	B	D	B	B	B	C	E

الخاتمة

الحمد لله الذي بنعمته تتم الصالحات، هذا وما كان من توفيق وسداد فمن الله وما كان من خطأ أو زلل أو نسيان فمن أنفسنا ومن الشيطان، وفقنا الله وإياكم لخيري الدنيا والآخرة وأعاننا وإياكم على ذكره وشكره وحسن عبادته ، تذكروا جميع من سبقنا من إخواننا بدعوة صادقة لعل الله انه يرفع قدرهم ويوسع مدخلهم ويسكنهم الفردوس الأعلى وأن يجعل الله هذا العمل في ميزان حسناتهم .